Keep, Measure, Erase

What a Quantum Loop Does with Its History

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Quantum loops have a reputation for paradox. We trace the paradoxes to a single fact: the history of a quantum loop, the record of how many times its body ran, is itself quantum information. A loop can erase its history, keep it, or measure it, and these choices give three different programs. Each competing account of quantum loops makes one of these choices implicitly. We make the choice explicit in iso, a reversible language in the Π family, and the paradoxes dissolve. The coherent loop, which erases its history, is the trace of its body: for every finite body it is total and unitary and compiles exactly into loop-free code, so the question of when it halts disappears. The other two loops come from the coherent one through two universal constructions on quantum processes: R, which adds allocation, yields the loop that keeps its history, and L, which then adds hiding, yields the loop that measures it. Measuring is therefore keeping and then forgetting, and forgetting is harmless exactly when the history is independent of the data, which a compiler can decide. Finally, the coherent loop is physical: light circling an optical ring resonator computes it. This gives the trace a physical meaning that recent work had left open, and it opens a new front for quantum programming: loops that run as optical feedback.

1 Introduction

Can a quantum computer run a loop? The question sounds settled, yet the answers on record contradict one another. Let the loop's guard be in superposition, so that one branch exits after three rounds while another keeps going. One answer says the computer then cannot halt without spoiling its computation [17]. A second says the loop must measure its guard on every round, which leaves a classical loop running over quantum data [26]. A third keeps the guard coherent but bounds the rounds and pads every branch until all of them finish together [27, 28]. A fourth defines the loop as a trace, an infinite sum over the paths of its control flow [3, 4]. We claim that all four answers are right, each about a different loop. A quantum loop consists of a body and a history, the record of how many times the body ran, and the four answers differ only in what happens to that history.

Physics has a loop that nobody counts. An ideal lossless ring resonator has a straight waveguide and a ring that meet at a single coupler. Light entering the coupler splits: part continues straight through, and part goes around the ring and meets the coupler again, where it splits again. Some light leaves after one trip, some after two, some after a hundred. Under steady illumination at a single frequency, all these contributions reach the same outgoing mode, so their amplitudes add [2, Eq. (6)]. The ring thus gives a physical example of the execution formula from geometry of interaction: its output adds the amplitude that goes straight through to the amplitudes that leave after each trip around the ring [1, 4].

In Q# and OpenQASM 3, a `while` loop tests a classical condition [19, 20]. If we measure the guard after each round to decide whether to continue, the measurement distinguishes an exit on this round from an exit on a later one. Keeping the guard coherent brings us back to a question about halting on a quantum Turing machine [17]. If different branches finish at different times, when can we read the output without learning which branch finished first?

We can see how the time of exit becomes a record by changing the light we send into the same ring. Suppose we send in a pulse that lasts less than the time it takes light to go around the ring once. The part that goes straight through has already passed the output before the part that went around once arrives. Each further trip gives another pulse, one round-trip time later. The arrival time now records the count [21, Eqs. (30)–(33)].

A quantum language has three choices for what to do with a loop's history:

- **Keep it.** Return a count register with the data.
- **Measure it.** Measure the guard after each call to decide whether to continue.
- **Erase it.** Add the amplitudes from different rounds at the same output, as the ring does.

For classical programs the three coincide on the returned data when the count is ignored, which is why classical languages never had to choose. For quantum programs they give three different loop constructs, and a language that provides only one has chosen for its programmers.

We make this choice explicit in iso, a reversible language in the Π family. We write the body once using symmetric pattern-matching clauses [22, §§ 3–5], with an exit and a back edge, and close it in three ways. The coherent loop `trace` connects the back edge, count_K records the number of returns modulo K, and `while` measures the guard after each call. Each choice comes with something a compiler can use:

- **Coherent loops compile away.** Starting from the body's table of clauses, we eliminate the values on the back edge one at a time. This gives a finite iso with exact coefficients and no back edge. There is one elimination step per returning basis value, even when infinitely many rounds contribute [3, Thm. 3.3.15].
- **Forgetting can be certified.** Each possible return count gives an exit map: on a given input, it gives the amplitude that leaves after exactly that many returns. Suppose the back edge has m basis values and the external data type is nonempty. We check whether the first $m + 1$ exit maps are all scalar multiples of the same map. With exact coefficients whose equality we can decide, this test tells us when the measured loop may replace the coherent one in every well-typed channel context. That includes contexts where the input is entangled with a reference the caller keeps.
- **We can probe the coherent loop without a counter.** Put a phase on the back edge. Amplitude that goes around once picks up the phase once, and amplitude that goes around twice picks it up twice. How the output changes as we vary the phase tells us how much weight circulates on the back edge. One extra control qubit lets us compare two nearby phases, and repeated measurements let us estimate that weight. Averaging the weight over the phase gives the measured loop's mean number of returns. For the rotation with $\cos\theta = 3/5$ and $\sin\theta = 4/5$, the measured loop makes one return on average. With no added phase, the coherent loop circulates four times the input weight and returns the input with its sign flipped.
- **An ideal ring realizes the scalar loop.** A lossless coupler can give the same amplitude as our scalar `trace` for each trip around the ring. Its response at a single frequency is therefore the same coherent sum. When the coupler is independent of frequency and each trip adds a phase proportional to frequency, multiplying the circulating weight for a unit input by the round-trip time gives the ring's Wigner–Smith delay [23].

2 Finite programs in iso

An iso $\omega : a \leftrightarrow b$ is a reversible program from data type a to data type b. We use the finite types

0	1	$a + b$	$a \otimes b$
no values	one value $*$	a tagged value from a or b	a pair from a and b.

These are the sum and product types of Π, including the empty and unit types [11, § 3.1]. Write $\mathcal{B}(a)$ for the basis values of a and $|s\rangle$ for the basis vector of $s \in \mathcal{B}(a)$. For example, $\mathcal{B}(1) = \{*\}$ and $1 + 1$ has values left $*$ and right $*$.

2.1 Syntax

An iso can be given by a table of clauses. To read a clause $v \leftrightarrow e$, match the input against v and use the variables bound by the match to evaluate e. For example, the pattern left x matches left s

and binds x to the payload s. A pattern may be unit $*$, a variable x, a tagged pattern left v or right v, or a pair $\langle v_1, v_2 \rangle$. Each variable occurs at most once in the pattern. The grammar also allows an expression to form a finite superposition with coefficients $\alpha_i \in \mathbb{C}$, or to apply an iso locally:

$$e ::= v \mid \textstyle\sum_i \alpha_i e_i \mid \mathsf{let}\ p = \omega\ q\ \mathsf{in}\ e,$$
$$\mathcal{T} ::= \{v_i \leftrightarrow e_i\}_{i \in I}, \qquad t ::= e \mid \omega\ t,$$
$$\omega ::= f \mid \mathcal{T} \mid \omega ; \omega \mid \omega \oplus \omega \mid \omega \otimes \omega \mid \omega^{\dagger}.$$

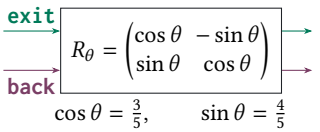
Here I is finite. In a local application, q assembles the argument to ω, and p binds the result for the rest of the expression. Both p and q use only unit, variables and pairs. A variable can bind a value of any type, including a whole tagged value. A pair pattern takes apart a result of product type. In this way the same binding works for every basis value that the local iso can return.

An iso may be a clause table, a name f for an earlier checked definition, or a combination of isos. Sequence $f ; g$ runs f first and then g. A sum acts on the alternative selected by the tag, a tensor acts on the two factors of a pair, and dagger reverses the iso. A closed term t has no free value variables. We write application $\omega\ t$ as `omega on t` in the executable syntax. Thus `f ; g on t` applies the whole sequence to t: it feeds the result of f directly into g.

In general, **`round`** $f : a + c = \mathcal{T}$ names a table of clauses $\mathcal{T}$ and gives the open body f type $(a + c) \leftrightarrow (a + c)$. This abbreviation uses the same external type a on both sides. The outer tags **`exit`** and **`back`** stand for **`left`** and **`right`** and carry values of type a and c, respectively. The payload can itself be tagged. For example, **`Qubit`** is the executable name for $1 + 1$, with `0` for left $*$ and `1` for right $*$. Thus `exit 0` expands to left (left $*$): the outer tag selects the port and the inner tag gives the qubit value. Similarly, **`Unit`** names the type 1 and `unit` its value $*$.

For the loop from the introduction, take both payload types to be **`Unit`**. The guard then has just two basis values, `exit unit` and `back unit`. We want one call to rotate this guard through $\theta = \arccos(3/5)$. With the exit basis value first and the return basis value second, the two clauses give the two columns of the rotation matrix shown beside them.

```
round guardRound : Unit + Unit = {
  exit unit ↔ [ 3/5 . exit unit + 4/5 . back unit];
  back unit ↔ [-4/5 . exit unit + 3/5 . back unit]
}
```



On input `exit unit`, the first clause gives $(3/5)|\mathsf{exit}\ *\rangle + (4/5)|\mathsf{back}\ *\rangle$. The brackets group the superposition, and the dot scales each tagged value by its amplitude. These two amplitudes form the first column of R_θ. The `back unit` clause gives the second column in the same order.

The closing brace completes the definition of one call. In the diagram, amplitude enters from outside through the upper input and leaves through the upper output. To run another round on the returning amplitude, we connect the lower output to the lower input, as we do in Section 3.

To evaluate a source iso ω, we unfold its names and replace each checked clause table by an atom $[M]$ containing that table's matrix. The result is a program κ built from *matrix combinators*:

$$\kappa ::= [M] \mid \kappa ; \kappa \mid \kappa \oplus \kappa \mid \kappa \otimes \kappa \mid \kappa^{\dagger}.$$

For `guardRound` the whole table becomes the single atom $\begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}$. Before a table can become a matrix atom, its columns must pass the unitary check.

2.2 Statics

To check a table $\omega : a \leftrightarrow b$, we first check that each basis input of a matches exactly one clause and that each output expression has type b under the variables its pattern binds. Evaluating that clause on s gives column s of $M = [\![\omega]\!]$. Its entry in row t is the amplitude for output t. For `guardRound`, the marked entry $M_{ts} = -4/5$ is the exit amplitude on a returning input.

	exit	s=back
t=exit	3/5	−4/5
back	4/5	3/5

Both columns of `guardRound` have squared norm one, and their inner product is zero: the exit row contributes $-12/25$ and the return row contributes $12/25$ (Figure 1). In general, we check that a and b have equally many basis values and that $M^\dagger M = I$, where dagger takes the conjugate transpose. The diagonal entries of this product are the squared column norms and the off-diagonal entries are the inner products of distinct columns. A square matrix that passes this check is unitary.

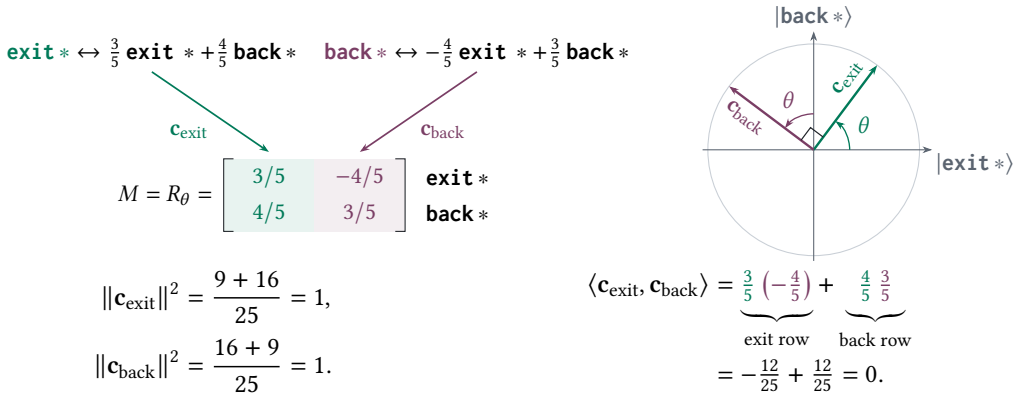


Fig. 1. Each clause gives one column of R_θ. The rotation turns each basis vector through the same angle θ, so the columns lie on the unit circle and remain perpendicular. Each product in their inner product uses entries from the same output row. For complex columns we conjugate the first entry of each product.

The judgment $\Sigma; \Delta \vdash e : b$ gives e output type b in the earlier iso definitions Σ and the context of value variables Δ. These variables are used linearly. In a pair, the two components use disjoint parts of Δ. In a superposition, each alternative uses the same variables once. A local application splits the variables between its argument and its continuation, where the result pattern binds fresh variables.

A checked iso has judgment $\Sigma \vdash \omega : a \leftrightarrow b$. Matrix atoms pass the same column test as tables, and the combinators must agree on the types they connect. A named declaration enters Σ only after it passes these checks, so evaluating a clause during checking can use only earlier definitions. Within a definition, checking and evaluation recurse on proper subprograms. The clauses and linear contexts adapt symmetric pattern matching [22, §§ 3–5] [16, Ch. 3]. Appendix A gives the full rules.

2.3 Dynamics

In Figure 1, we obtained each column by evaluating the right-hand side of a clause. Write $\gamma \vdash e \Downarrow \psi$ when expression e evaluates to the vector ψ under the basis values assigned by γ. A value v gives $|v[\gamma]\rangle$, and a sum adds its weighted outputs. A local application passes each basis result, with its amplitude, to the continuation and adds the resulting vectors.

Write $\mathcal{E}_\Sigma(\omega) = \kappa$ for elaboration and $\Sigma \vdash \omega \Downarrow M$ when κ evaluates to M. For `guardRound`, the atom rule $[R_\theta] \Downarrow R_\theta$ gives the matrix whose columns we just checked. A closed term then applies the matrix to its input: if $t \Downarrow \psi$, then $\omega\, t \Downarrow M\psi$.

Suppose we apply guardRound twice in sequence, starting with `exit unit`. After the first call the amplitudes are $3/5$ on **exit** and $4/5$ on **back**. The second call therefore uses both clauses. Its output is the first column weighted by $3/5$ plus the second column weighted by $4/5$:

$$\frac{3}{5}\begin{pmatrix}3/5\\4/5\end{pmatrix}+\frac{4}{5}\begin{pmatrix}-4/5\\3/5\end{pmatrix}=\begin{pmatrix}-7/25\\24/25\end{pmatrix}.$$

This is ordinary sequence: the whole output of the first call becomes the input to the second. In general, if $\kappa \Downarrow M$ and $\lambda \Downarrow N$, then $\kappa;\lambda \Downarrow NM$, since M acts first on the input column. Elaboration preserves sequence, sum, tensor and dagger; the last three evaluate by taking the direct sum, tensor product and conjugate transpose of the corresponding matrices.

2.4 Denotation

A type a denotes the space $[\![a]\!] = \mathbb{C}^{\mathcal{B}(a)}$, with one orthonormal basis vector for each basis value of a. Thus 0 and 1 denote the zero space and $\mathbb{C}$, and sum and product types denote direct sums and tensor products, respectively.

A checked iso $\omega : a \leftrightarrow b$ denotes the linear map from $[\![a]\!]$ to $[\![b]\!]$ determined by its evaluated columns. For guardRound, the two clauses tell us what happens to the basis inputs `exit unit` and `back unit`. Now suppose the input has amplitude u on **exit** and v on **back**. By linearity, one call gives the first column weighted by u plus the second column weighted by v:

$$[\![\mathsf{guardRound}]\!]\begin{pmatrix}u\\v\end{pmatrix}=u\begin{pmatrix}3/5\\4/5\end{pmatrix}+v\begin{pmatrix}-4/5\\3/5\end{pmatrix}=\begin{pmatrix}(3u-4v)/5\\(4u+3v)/5\end{pmatrix}.$$

This is what it means for guardRound to denote R_θ: multiplying by R_θ gives the output of the two clauses on every input. The column check makes this map unitary. Appendix A proves the correspondence with evaluation for all checked isos.

In the sequence above, we passed both output coordinates to the next call. To form a loop, we let the **exit** amplitude leave and feed only the **back** amplitude into the next call.

3 What the coherent loop computes

Closing guardRound leaves the caller with a single output of type **Unit**. Every amplitude that leaves through **exit** contributes to that output, including amplitude that returns many times before it leaves.

3.1 From a round to its exits

Follow the lower wire in Figure 2. On the second call, only the **back** clause contributes: it takes the returning amplitude $4/5$ and sends $(4/5)(-4/5) = -16/25$ to the external output. The rest goes around again.

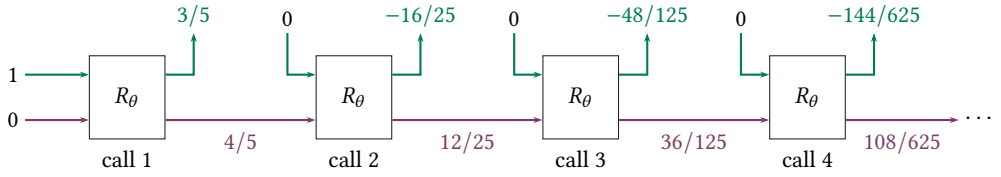


Fig. 2. Four calls to guardRound. The upper outputs contribute to the caller's result; the first two add to $-1/25$. The lower wire carries the returning amplitude from one call to the next. A zero at an input means no amplitude enters that port.

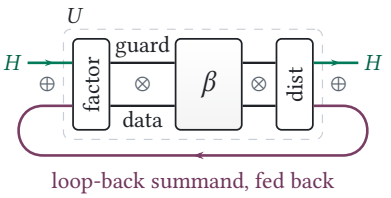
Syntax and statics. The programmer writes `trace guardRound`. Unfolding the name and replacing its two clauses by their checked matrix atom gives the evaluator $\mathrm{trace}_1[R_\theta]$. Both programs connect the same return wire. The two typing rules describe this connection before and after elaboration:

$$\underbrace{\frac{\Sigma \vdash \omega : a + c \leftrightarrow b + c}{\Sigma \vdash \mathrm{trace}_c\, \omega : a \leftrightarrow b}}_{\text{source iso}} \qquad \underbrace{\frac{\kappa : a + c \leftrightarrow b + c}{\mathrm{trace}_c\, \kappa : a \leftrightarrow b}}_{\text{matrix combinator}}.$$

In the source rule, Σ supplies the earlier definitions that ω may name. In the matrix rule those names have already been unfolded. In each premise, the output and input to be connected have the same type c. In each conclusion, only the external input a and output b remain. Thus `trace guardRound` has type $1 \leftrightarrow 1$; the **round** declaration supplies the return type 1. In general, if $\mathcal{E}_\Sigma(\omega) = \kappa$, then $\mathcal{E}_\Sigma(\mathrm{trace}_c\, \omega) = \mathrm{trace}_c\, \kappa$. Both grammars gain one constructor, $\omega ::= \cdots \mid \mathrm{trace}_c\, \omega$ and $\kappa ::= \cdots \mid \mathrm{trace}_c\, \kappa$. In `trace f; g`, we close `f` first, then apply `g`. QWhile uses this typed feedback interface for its do-while combinator [3, § 5.1].

Elaboration preserves how the program connects its parts. We can evaluate the resulting combinator to a matrix or draw those same connections as a diagram. A box for an iso denotes a morphism in Unitary, the category of finite-dimensional Hilbert spaces and unitary maps. Joining boxes in sequence composes their maps. Closing a return wire takes the trace for direct sum [4, 14]. Suppose both tags carry a payload H, and write $G = \mathbb{C}^2$ for the guard. Read the picture from left to right. Its factor box is the unitary $H \oplus H \to G \otimes H$ that sends **exit** x to $|0\rangle \otimes x$ and **back** x to $|1\rangle \otimes x$. This includes the swap that puts the guard first. The body β acts from $G \otimes H$ to itself, and the dist box is the inverse of the factor box. Closing the lower H leaves a map $H \to H$ for the caller. The general rule also allows different payload types.

$\mathrm{trace}_H(\mathrm{factor}; \beta; \mathrm{dist})$



The exit maps. To follow the same calculation for any body $\beta : a + c \leftrightarrow b + c$, evaluate it to a matrix U and group its columns by the input tag and its rows by the output tag. Write $X = [\![a]\!]$, $X' = [\![b]\!]$ and $Y = [\![c]\!]$. On external input $u \in X$ and returning input $y \in Y$, one call gives

$$U \begin{pmatrix} u \\ y \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} u \\ y \end{pmatrix} = \begin{pmatrix} Au + By \\ Cu + Dy \end{pmatrix}. \tag{3.1}$$

The top row gives the external output in X' and the bottom row gives the returning output in Y. For `guardRound`, the first clause gives $A = 3/5$ and $C = 4/5$, and the second gives $B = -4/5$ and $D = 3/5$. Reading the same rows and columns gives $A : X \to X'$, $B : Y \to X'$, $C : X \to Y$ and $D : Y \to Y$. These are morphisms in **FdHilb**, where all linear maps are allowed. Since U itself is unitary and the same finite space Y occurs on both sides, $\dim X = \dim X'$.

The first call receives $(u, 0)$, so Au leaves immediately and Cu goes back. The second call receives $(0, Cu)$, so BCu leaves and DCu goes back. These are components of the original input amplitude, with no normalization between calls. The zero denotes a port with no incoming amplitude. Taking either component of the output is a projection in **FdHilb**.

Write $x_k : X \to X'$ for the exit map after k returns. The immediate exit has made zero returns. To exit after $k + 1$ returns, a contribution first enters Y through C, makes k further passages through D, and finally leaves through B. Thus

$$x_0 = A, \qquad x_{k+1} = BD^k C. \tag{3.2}$$

After $n+1$ calls the contributions to the external output are $x_0u, \dots, x_nu$, with D^nCu still on the return wire. Coherent feedback adds all the exits at their common output:

$$[\![\operatorname{trace}_c \beta]\!] = L = \sum_{k\geq 0} x_k.$$

This is the execution formula from the introduction [1]. In the ring, x_0 gives the light that goes straight through and x_k gives the contribution after k trips around the ring. A finite unitary body can have infinitely many nonzero exits. To compute their sum, we follow the amplitude on the return wire.

3.2 Computing the sum without a last round

Follow the return wire of `guardRound` in Figure 2. On input `unit`, the first call sends $4/5$ back. Each later call returns $3/5$ of the amplitude it received on that wire. The four labels on the lower wire are therefore the first four terms of the same geometric series. Write y for its sum. Since each term is $3/5$ times the preceding one, the series converges and its tail is $3/5$ times the whole sum:

$$y = \frac{4}{5} + \underbrace{\frac{12}{25} + \frac{36}{125} + \frac{108}{625} + \frac{324}{3125} + \cdots}_{(3/5)\,y} = \frac{4}{5} + \frac{3}{5}y.$$

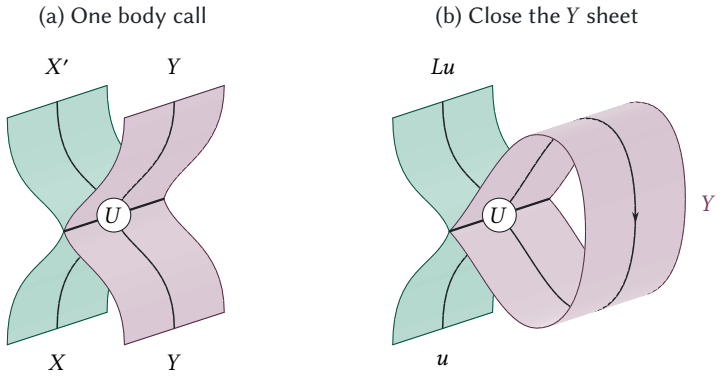
This is the *wire equation*. Subtracting $(3/5)y$ from both sides gives $(2/5)y = 4/5$, so the returns sum to $y = 2$. On the external wire, the first call contributes $3/5$ directly. Each returning amplitude contributes to that same output with coefficient $-4/5$. Their sum is $3/5 - (4/5)y = -1$, so `trace guardRound on unit` $\Downarrow -|*\rangle$.

For a general body, the same argument starts with Cu on the return wire and applies D on each further passage. For a finite unitary body, the external input reaches only the part of the return space where these contributions decay geometrically (Theorem B.1 and its proof). Separating the first return and then collecting the exits gives

$$y = Cu + DCu + D^2Cu + \cdots = Cu + Dy,$$
$$Lu = Au + B(Cu + DCu + D^2Cu + \cdots) = Au + By.$$

For finite unitary bodies the exits are absolutely summable, L is unitary, and $\sum_k x_k^\dagger x_k = I_X$ [4, Thms. 4, 7, 8] [3, Thm. 3.3.15, Cor. 3.3.16].

The first equation also follows directly from the connection: the vector $Cu + Dy$ leaving the returning port is the vector y entering it. Figure 3 draws this connection as a sheet diagram, read upwards. Separate sheets carry sum alternatives; wires at different depths on one sheet carry tensor factors [8, Defs. 16, 18–20]. The circled U acts on both incident summands, including the one partly hidden by the front sheet.



For `guardRound` on input `unit`, the wire equation gives

$$R_\theta \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

The purple output matches the purple input. The caller receives the green output, -1.

Fig. 3. In Unitary, the open body at left is $U : X \oplus Y \to X' \oplus Y$. Joining the two ends of the purple Y sheet at right gives its direct-sum trace $L = \mathrm{Tr}^Y(U) : X \to X'$.

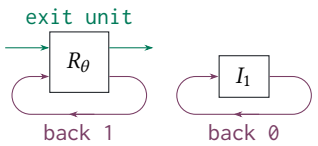
For `guardRound`, one basis input gave the whole closed matrix. For a larger input type, the same wire equation gives one column at a time.

Write F for the matrix whose column s holds the sum of the returns on input $|s\rangle$. The first call contributes $C|s\rangle$ to that column. Every later call takes what the previous call sent back and applies D, so the later calls together contribute $DF|s\rangle$. Writing this for all inputs gives $F = C + DF$, or $(I_Y - D)F = C$. At the external output, A gives the immediate exits and B sends the accumulated returns to that same output. To evaluate the loop, we evaluate the open body κ, find a solution F of the wire equation, and return the gate $A + BF$.

$$\frac{\kappa \Downarrow U \quad U = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \quad F : X \to Y \quad (I_Y - D)F = C}{\mathrm{trace}_c\, \kappa \Downarrow A + BF}.$$

An undetermined return. For `guardRound` we found the return by dividing by $1 - 3/5$. To see what happens when $I_Y - D$ has no inverse, give the body two return values. Use `back 1` for the original return and let the new value, `back 0`, return to itself.

```
round extraPort : Unit + Qubit = {
  exit unit ↔ [ 3/5 . exit unit + 4/5 . back 1];
  back 1    ↔ [-4/5 . exit unit + 3/5 . back 1];
  back 0    ↔ back 0
}
```



The first two clauses are the rotation we already know. Only the last clause mentions `back 0`. In the order `exit unit`, `back 1`, `back 0`, the body therefore has matrix $R_\theta \oplus I_1$. On external input u, write y_1 and y_0 for their respective return amplitudes. The clauses give

$$y_1 = \tfrac{4}{5}u + \tfrac{3}{5}y_1, \qquad y_0 = y_0.$$

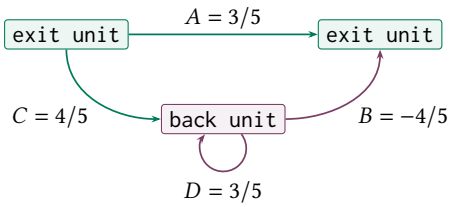
The first equation fixes $y_1 = 2u$. The second holds for every y_0. This is why $I_Y - D = \mathrm{diag}(2/5, 0)$ is *singular*, meaning it has no inverse. Starting from u, no call sends any amplitude to `back 0`, so the actual returns there sum to zero. Even if the evaluator chooses another y_0, only y_1 contributes to the exit. The caller still receives $(3/5)u - (4/5)y_1 = -u$.

For any finite unitary body, the external output is fixed even when the return is undetermined. The sum of the returns supplies a solution on every external input. If two solutions differ by w, subtracting their wire equations gives $Dw = w$. On input $(0, w)$ the body therefore returns w with

its full norm. Unitarity forces the external component Bw to be zero, so both solutions give the same column of L. The full argument is in Appendix B, Figure 16.

3.3 Compiling away the loop

The wire equation gave us the matrix $L = A + BF$ of the closed loop. We can also compute this gate directly in the clause table. In `guardRound`, the first clause sends $3/5$ to `exit unit` and $4/5$ to `back unit`. To remove the back edge, we need to know what that second term will contribute to the output. Follow the arrows from `back unit`. One call sends $B = -4/5$ to the exit and $D = 3/5$ back to the same clause.



Each trip around the self-loop contributes another factor of D. From one unit of returning amplitude, the exits therefore sum to

$$B + BD + BD^2 + \cdots = \frac{B}{1 - D} = -2.$$

The first clause sends only $C = 4/5$ there, so these exits contribute $(4/5)(-2)$. On the wire we grouped the same factors as $BF = (-4/5) \cdot 2$, with $F = C/(1 - D)$.

We can now replace `back unit` in the first clause by the sum of its exits, -2 (`exit unit`). The clause becomes

$$\texttt{exit unit} \;\leftrightarrow\; \tfrac{3}{5}\,(\texttt{exit unit}) + \tfrac{4}{5}\big(-2\,(\texttt{exit unit})\big) = -\texttt{exit unit}.$$

Both terms leave through `exit unit`, so their coefficients add. Dropping the external tag gives $\{* \leftrightarrow -1 \cdot *\}$.

With two return patterns, removing the first can change the clause we must use for the second. In `flipRound`, each call rotates the same guard and flips a data bit (Appendix F.1). The `back 0` clause sends $-4/5$ to `exit 1` and $3/5$ to `back 1`. Figure 4 substitutes this expression in the two clauses that mention `back 0`.

This creates a return from `back 1` to itself with coefficient $(3/5)^2 = 9/25$. We close that return before substituting in the external clauses.

For a general body, first evaluate the matching clause on each basis input, as in Section 2. In the resulting table, choose a returning basis value **back** q and write its clause as **back** $q \mapsto d\,(\textbf{back}\,q) + e$. Here $d = D_{qq}$ is the amplitude to return to q and e contains the other outputs. If $d \neq 1$, substitute $e/(1 - d)$ for **back** q in every other clause (Appendix B.1). Thus $v \mapsto e' + \alpha\,(\textbf{back}\,q)$ becomes $v \mapsto e' + \alpha e/(1 - d)$. Collect terms on equal patterns and delete the clause for q. If $d = 1$, unitarity forces all other entries in its row and column to vanish. We can simply delete that clause, as for `back 0` in `extraPort`.

Each step leaves a smaller unitary table whose coefficients we use for the next step. After $|\mathcal{B}(c)|$ steps, dropping the external tags gives an ordinary iso of type $a \leftrightarrow b$ with matrix L. This applies the elimination for finite contractions to a unitary clause table [3, Thm. 3.3.15]. Proposition B.2 shows that every order gives the same map and keeps the coefficients in their original subfield.

Remove back `0`. Replace it by $-\frac{4}{5}\mathtt{exit}\,1 + \frac{3}{5}\mathtt{back}\,1$.

$$\begin{array}{c|c} \mathtt{exit}\,1 \mapsto \frac{3}{5}\mathtt{exit}\,0 + \frac{4}{5}\mathtt{back}\,0 & \mathtt{back}\,1 \mapsto -\frac{4}{5}\mathtt{exit}\,0 + \frac{3}{5}\mathtt{back}\,0 \\ \Downarrow & \Downarrow \\ \frac{3}{5}\mathtt{exit}\,0 + \frac{4}{5}\left(-\frac{4}{5}\mathtt{exit}\,1 + \frac{3}{5}\mathtt{back}\,1\right) & -\frac{4}{5}\mathtt{exit}\,0 + \frac{3}{5}\left(-\frac{4}{5}\mathtt{exit}\,1 + \frac{3}{5}\mathtt{back}\,1\right) \\ = \frac{3}{5}\mathtt{exit}\,0 - \frac{16}{25}\mathtt{exit}\,1 + \frac{12}{25}\mathtt{back}\,1 & = -\frac{4}{5}\mathtt{exit}\,0 - \frac{12}{25}\mathtt{exit}\,1 + \frac{9}{25}\mathtt{back}\,1 \end{array}$$

Delete the back `0` clause. The third clause is $\mathtt{exit}\,0 \mapsto \frac{3}{5}\mathtt{exit}\,1 + \frac{4}{5}\mathtt{back}\,1$.

Close back `1`. Sum its exits with factor $1/(1 - 9/25) = 25/16$.

$$\mathtt{back}\,1 \;\longmapsto\; \tfrac{25}{16}\left(-\tfrac{4}{5}\mathtt{exit}\,0 - \tfrac{12}{25}\mathtt{exit}\,1\right) = -\tfrac{5}{4}\mathtt{exit}\,0 - \tfrac{3}{4}\mathtt{exit}\,1.$$

Substitute for back `1` **in both exit clauses.**

$$\mathtt{exit}\,0 \mapsto \tfrac{3}{5}\mathtt{exit}\,1 + \tfrac{4}{5}\left(-\tfrac{5}{4}\mathtt{exit}\,0 - \tfrac{3}{4}\mathtt{exit}\,1\right) = -\mathtt{exit}\,0,$$

$$\mathtt{exit}\,1 \mapsto \tfrac{3}{5}\mathtt{exit}\,0 - \tfrac{16}{25}\mathtt{exit}\,1 + \tfrac{12}{25}\left(-\tfrac{5}{4}\mathtt{exit}\,0 - \tfrac{3}{4}\mathtt{exit}\,1\right) = -\mathtt{exit}\,1.$$

Fig. 4. Compiling `trace flipRound`. The arrows substitute for back `0`. Closing back `1` and substituting its exits leaves each input bit with amplitude -1.

3.4 Making the sign observable

The clause for `trace guardRound` sends $|*\rangle$ to $-|*\rangle$. A caller who measures the result still sees $*$ with certainty, just as for the identity. To see the sign, the caller needs a second amplitude to compare with the one that went through the loop. In `bypassRound`, the two amplitudes belong to one qubit. Figure 5 draws them as two paths. The upper path carries $|0\rangle$ straight through. The lower path carries $|1\rangle$ through `guardRound`. The return still carries **`Unit`**.

```
round bypassRound : Qubit + Unit = {
  exit 0 ↔ exit 0;
  exit 1 ↔ [ 3/5 . exit 1 + 4/5 . back unit];
  back unit ↔ [-4/5 . exit 1 + 3/5 . back unit]
}
```

The first clause sends all of `exit 0` straight through, so the bypass appears only in the immediate exit x_0. Every later exit comes from `back unit`, whose only external output is `exit 1`. Read the clauses as columns in the order `exit 0`, `exit 1`, `back unit`, with $t = 3/5$ and $s = 4/5$:

$$U_{\text{bypass}} = \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & t & -s \\ \hline 0 & s & t \end{array}\right), \qquad x_0 = \operatorname{diag}(1, t), \qquad x_{k+1} = \operatorname{diag}(0, -s^2 t^k) \quad (k \geq 0).$$

On the lower path, a later exit picks up s on entering the return wire, t on each further return, and $-s$ on leaving. These are the same exits we summed for `guardRound`. Closing the loop therefore leaves $|0\rangle$ alone and sends $|1\rangle$ to $-|1\rangle$.

The caller in Figure 5 runs the Hadamard test on the scalar loop [7, p. 1, Eqs. (1)–(3)]. Its program is `H; trace bypassRound; H; measure on 0`. The first Hadamard prepares $(|0\rangle + |1\rangle)/\sqrt{2}$. The loop changes this to $(|0\rangle - |1\rangle)/\sqrt{2}$. Measuring either state in the computational basis would give 0 and 1 with equal probability. The second Hadamard lets both paths contribute to each output before

the caller measures.

$$H\frac{|0\rangle - |1\rangle}{\sqrt{2}} = \frac{1}{2}(|0\rangle + |1\rangle) - \frac{1}{2}(|0\rangle - |1\rangle)$$
$$= \underbrace{\left(\tfrac{1}{2} - \tfrac{1}{2}\right)}_{\text{output 0}}|0\rangle + \underbrace{\left(\tfrac{1}{2} + \tfrac{1}{2}\right)}_{\text{output 1}}|1\rangle = |1\rangle.$$

The contributions cancel at output 0 and add at output 1, so the caller reads 1 with certainty. Had the loop preserved the sign, the two Hadamards would have returned the caller to 0.

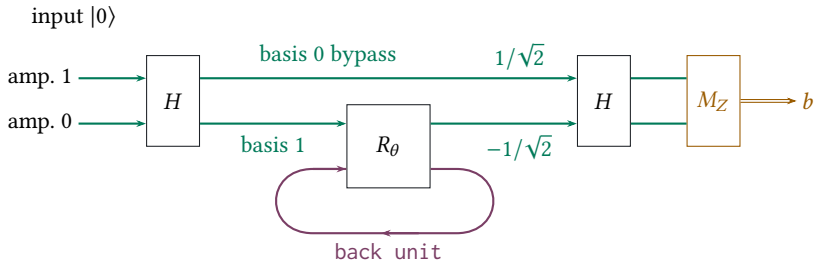


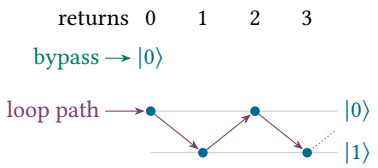
Fig. 5. The two rails carry the basis alternatives of one qubit $G = \mathbb{C}^2$. Each $H : G \to G$ and the closed loop $\mathrm{diag}(1, -1) : G \to G$ are in Unitary. The final measurement $M_Z : G \rightsquigarrow G$ is a channel in CPTP. Its output is diagonal in the computational basis and gives bit 1 with certainty.

The two paths in Figure 5 also have a use in optics. A splitter sends light along two waveguides. One takes it straight to a second splitter. The other meets the ring from the introduction. Some light goes straight through, and some makes one or more trips around the ring before rejoining that waveguide. The second splitter brings the two paths together. How the waves line up there determines which output gets more light. Wang et al. used this to sort light by wavelength. They chose the path lengths so that neighboring wavelength channels went mostly to different outputs [24, Figs. 1 and 6].

In our program, output 0 never occurs because the bypass amplitude cancels the sum of the exits from `guardRound`. Suppose we add a counter to find out how many times the loop returned. A classical programmer could keep this count for later without changing what the caller sees. Here a count can distinguish exits whose amplitudes we just added at the same output.

4 What allocation and hiding do to the loop

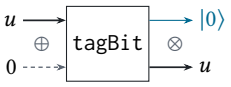
We run the same three clauses of `bypassRound`, this time with a counter bit. We start it at 0 and flip it on each return. The bypass exits at once with 0. On the lower path, one return takes the bit to 1 and two bring it back to 0. At the second Hadamard, exits with an even count can still interfere with the bypass. Odd counts carry a different bit.



The caller did not supply a count register. We need the ability to allocate that register.

Adding allocation to iso. Our caller supplies one qubit, with two basis values. Returning it alongside a counter bit gives four output pairs. An iso needs a square table, so we tag the input to give it four basis values as well.

```
tagBit :: (Qubit + Qubit) ↔ (Qubit x Qubit)
tagBit = {left x  ↔ ⟨0, x⟩;
          right x ↔ ⟨1, x⟩}
lift tagBit on 1
```



The bit records the input tag, so either clause can run backwards. The last line uses `lift` to supply the data through the left input. On `1` it gives `<0, 1>`. On the caller's superposition $u = (|0\rangle + |1\rangle)/\sqrt{2}$, it gives $|0\rangle \otimes u$. The 0 on the unused input means zero amplitude.

Following Heunen and Kaarsgaard, we call the extended language iso_a [11, § 3.3]. It embeds each iso with arr and adds lift:

$$\frac{\omega : a \leftrightarrow b}{\mathsf{arr}\,\omega : a \rightarrowtail b} \qquad \frac{\omega : a + d \leftrightarrow b}{\mathsf{lift}\,\omega : a \rightarrowtail b}.$$

If ω evaluates to U, then $\mathsf{arr}\,\omega$ uses the whole matrix. On input u, $\mathsf{lift}\,\omega$ gives $U(u, 0)$: only the columns for a are used. Those columns are still orthonormal, so the map preserves inner products. This is an *isometry*, written $\rightarrowtail$. Programs in iso_a compose in sequence, by direct sum and by tensor. They denote maps in the category of finite-dimensional isometries $\mathsf{Isometry} \simeq R[\mathsf{Unitary}]$ [11, Thm. 22].

Write $[K]$ for the sum of $K \geq 1$ copies of 1. The same construction gives alloc_j, which prepares the counter at j alongside the data. Starting at 0 and advancing on each return gives $\mathrm{count}_K\,\beta : a \rightarrowtail [K] \otimes b$ for a body $\beta : a + c \leftrightarrow b + c$. In a program we write `count K f on t`, with a positive numeral for `K`, the round for `f`, and its external input for `t`. The numeral specifies how many values the counter has.

Adding hiding to iso_a. The counting loop returns the counter and the data. Suppose we want to return just the data to the caller for the second Hadamard. We need the ability to hide the counter. To see what this does, group the exits of `bypassRound` by the bit they carry. Write u_0 for the sum of the data vectors at even counts, including the bypass, and u_1 for their sum at odd counts. The joint output is $|0\rangle \otimes u_0 + |1\rangle \otimes u_1$. If the caller keeps only the data, we take the partial trace over the bit. A unit vector w has density matrix $ww^\dagger$. For our joint output, grouping its rows and columns by the counter gives

$$\begin{pmatrix} u_0 u_0^\dagger & u_0 u_1^\dagger \\ u_1 u_0^\dagger & u_1 u_1^\dagger \end{pmatrix} \xrightarrow{\text{hide the bit}} u_0 u_0^\dagger + u_1 u_1^\dagger.$$

The partial trace adds the two diagonal blocks. The other two drop out because $\langle 0 \mid 1 \rangle = 0$. Exits carrying the same bit have already added their amplitudes in u_0 or u_1.

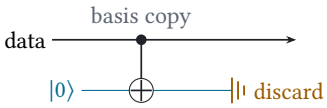
This second extension, iso_a^χ, embeds an isometry with arr and adds hide [11, § 3.4]:

$$\frac{v : a \rightarrowtail b}{\mathsf{arr}\,v : a \rightsquigarrow b} \qquad \frac{v : a \rightarrowtail b \otimes g \qquad \mathcal{B}(g) \neq \varnothing}{\mathsf{hide}\,v : a \rightsquigarrow b}.$$

The second rule hides the last factor, so for count_K we first exchange the factors in our counter-first formula. The executable `count` already returns the data first, so `count 2 bypassRound; discard` keeps just the data. The arrow $\rightsquigarrow$ denotes a channel. Its input state is a density operator ρ, a positive semidefinite matrix of trace one. If v evaluates to V, then $\mathsf{arr}\,v$ gives $V\rho V^\dagger$, and $\mathsf{hide}\,v$ gives $\mathrm{tr}_g(V\rho V^\dagger)$. Channel programs compose in sequence and by tensor. These channels are completely positive and trace-preserving. Huot and Staton identify their category as $\mathsf{CPTP} \simeq L[\mathsf{Isometry}]$ [12, Cor. 7]. Figure 14 gives the general diagrams for allocation and hiding.

To discard a whole register of inhabited type b, embed the iso $b \leftrightarrow 1 \otimes b$ and hide its last factor. The resulting channel $\mathrm{discard}_b : b \rightsquigarrow 1$ leaves just $\mathrm{tr}\,\rho$ at the unit output.

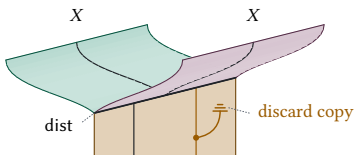
From a fresh bit to a measurement. Start a fresh bit at 0 and flip it when the data bit is 1. On basis input s this writes a copy of s. On a superposition $\alpha|0\rangle + \beta|1\rangle$, the two wires carry $\alpha|00\rangle + \beta|11\rangle$. Discarding the copy destroys the coherence between 0 and 1. Reading the data now gives 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$.



For an inhabited type b, lifting a permutation that sends each selected input s to $\langle s, s\rangle$ gives the same basis copier $\mathsf{clone}_b : b \rightarrowtail b \otimes b$. Hiding the copy gives computational-basis measurement [11, § 3.4.1]:

$$\mathsf{measure}_b = \mathsf{hide}\,\mathsf{clone}_b, \qquad [\![\mathsf{measure}_b]\!](\rho) = \sum_{s \in \mathcal{B}(b)} P_s \rho P_s, \qquad P_s = |s\rangle\langle s|.$$

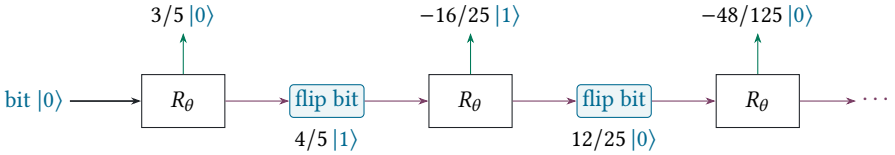
In the loop we copy only the **`exit`** or **`back`** tag. The picture shows the output of a call with common payload space X. Read upwards: the payload wire on the left passes through, and the tag wire on the right supplies the copy that ends at the orange discard mark. This measurement destroys coherence between the two tags and preserves it within each payload.



Appendix A.5 gives the rules for empty types. The counting loop carries one register across calls and returns it with the data. The measured loop $\mathsf{while}_c\,\beta : a \rightsquigarrow b$ discards a fresh copy of the tag after every call.

4.1 Keep a count

In `count 2 guardRound on unit`, the input is `unit` and the 2 tells us to keep the count modulo two. The first call sends 3/5 to the exit with the bit still at 0. The other 4/5 goes around with the bit flipped to 1, so the second exit carries 1. A second return flips it back to 0. The first and third exits now contribute to the same counter value. All three carry the unit payload.

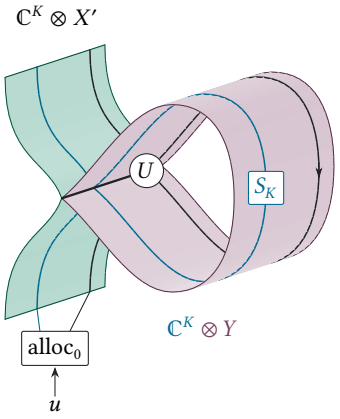


After the first return, every two further returns multiply the exit amplitude by $(3/5)^2$. Summing the exits with each bit value gives

$$\underbrace{\frac{3}{5} - \frac{48/125}{1-(3/5)^2}}_{\text{even returns}} = 0, \qquad \underbrace{\frac{-16/25}{1-(3/5)^2}}_{\text{odd returns}} = -1.$$

Thus the loop arm of `bypassRound` leaves with counter state $-|1\rangle$, while the bypass leaves with $|0\rangle$. On the caller's input $(|0\rangle + |1\rangle)/\sqrt{2}$, the joint output is $(|0\rangle \otimes |0\rangle - |1\rangle \otimes |1\rangle)/\sqrt{2}$, with the counter first. Hiding it leaves the data in the equal mixture $I/2$. The second Hadamard leaves this mixture unchanged, so the caller now reads 1 with probability 1/2.

For a counter with K values, replace the bit flip by the successor $S_K : j \mapsto j + 1 \bmod K$. Distributivity identifies $\mathbb{C}^K \otimes (X \oplus Y)$ with $(\mathbb{C}^K \otimes X) \oplus (\mathbb{C}^K \otimes Y)$, so each alternative sheet carries counter and payload together [8, § 6]. Let $d_K : [K] \otimes (a + c) \leftrightarrow ([K] \otimes a) + ([K] \otimes c)$ be distributivity, and let d'_K distribute over $b + c$. Figure 6 defines the enlarged body β_K and closes its return before supplying the allocated input. Its matrix W_K carries the counter on both alternatives and advances it only on a return.



The enlarged body and the count program are

$$\begin{aligned}\beta_K &= d_K^{-1}\,;\,(I_{[K]}\otimes\beta)\,;\,d'_K \\ &\quad;\,(I_{[K]\otimes b}\oplus(S_K\otimes I_c)),\\ \mathrm{count}_K\,\beta &= \mathrm{alloc}_0\,;\,\mathrm{arr}(\mathrm{trace}_{[K]\otimes c}\,\beta_K).\end{aligned}$$

Their matrices satisfy

$$\begin{aligned}W_K &= [\![\beta_K]\!] = \begin{pmatrix} I_K\otimes A & I_K\otimes B \\ S_K\otimes C & S_K\otimes D\end{pmatrix}.\\ R_K u &= (\mathrm{Tr}^{\mathbb{C}^K\otimes Y}\,W_K)(|0\rangle\otimes u)\\ &= \sum_{k\ge 0}|k \bmod K\rangle\otimes x_k u.\end{aligned}\tag{4.1}$$

Fig. 6. The count construction. Counter and payload are distinct tensor wires on both sheets, shown in depth in Figure 7(a). Only the returning counter passes through S_K.

The first call of W_K on $(|0\rangle\otimes u, 0)$ emits $|0\rangle\otimes Au$ and returns $|1 \bmod K\rangle\otimes Cu$. Here $|0\rangle$ is a prepared counter, while the bare 0 is absent amplitude. Closing $\mathbb{C}^K\otimes Y$ adds the exits, with the counter advanced once on each return. Figure 7 follows three calls with $K > 3$. The right panel shows where those same exits meet when we close the body without a counter.

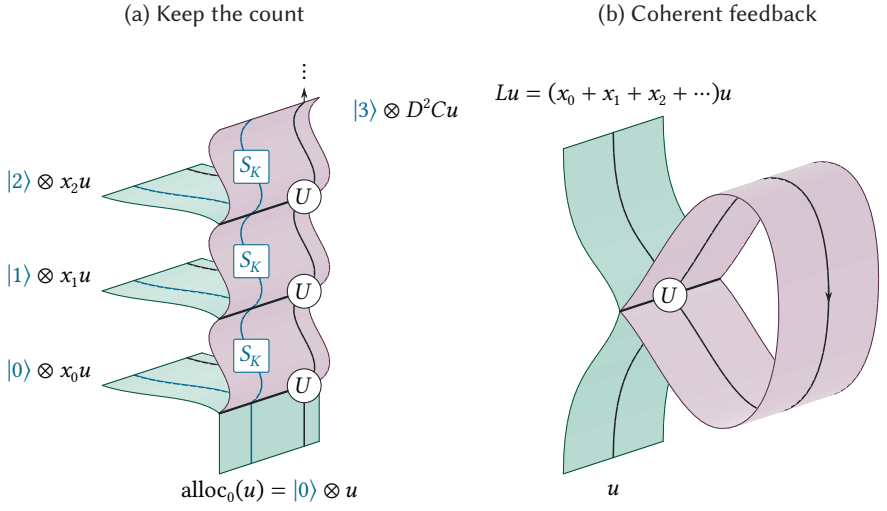


Fig. 7. After three calls of a $K > 3$ clock, $|3\rangle\otimes D^2Cu$ continues past the arrow and dots. The right panel closes the original body.

To hide the counter, first use the symmetry iso $\sigma_{[K],b} : [K]\otimes b \leftrightarrow b\otimes[K]$. Hiding then gives $\mathrm{hide}(\mathrm{count}_K\beta; \mathrm{arr}\,\sigma_{[K],b})$. Counts congruent modulo K share a label, so their amplitudes interfere before the clock is discarded. Hiding this finite clock therefore need not give the measured loop.

Allocation is an isometry and the enlarged coherent loop is unitary, so their composite R_K is an isometry. Evaluation returns the joint state without sampling a halting round.

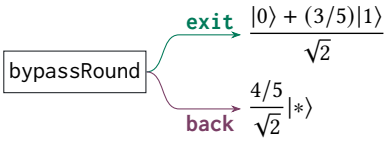
An exact count uses $H_{\mathrm{rec}} = \ell^2(\mathbb{N})$ and $Ru = \sum_k |k\rangle \otimes x_k u$, with $\|Ru\|^2 = \sum_k \|x_k u\|^2 = \|u\|^2$. It is the semantic limit of finite clocks. For the scalar rotation, $R|*\rangle = \xi_{\mathrm{rot}} \otimes |*\rangle$ with unit count state $\xi_{\mathrm{rot}} = t|0\rangle - s^2 \sum_{k\geq 1} t^{k-1}|k\rangle$. For `bypassRound`, the bypass instead has record $|0\rangle$: $R|0\rangle = |0\rangle \otimes |0\rangle$, while $R|1\rangle = \xi_{\mathrm{rot}} \otimes |1\rangle$. These different records of the two arms make $R|+\rangle$ entangled. If we embed each finite counter in H_{rec}, the maps R_K converge in operator norm to R as K grows (Appendix C). A time-dependent model can keep the same distinctions by making outputs from different calls arrive at different times [3, § 3.4].

In the bounded loop of Yuan et al., an input register says how many times to apply an operation. Their program runs for a fixed number of iterations. Once a branch has finished its work, it executes no-op instructions so that all branches reach the same control state at the same machine step [27, § 6.1, Fig. 7]. The original count remains in the input register. Register-machine compilation likewise computes wait counts within a chosen time bound [28, § 5] to address the common-clock halting question [17]. In our loop, amplitude from a single input can leave after different numbers of returns. The counter labels those exits as the body runs, even when there is no last round.

4.2 Measure the returning tag

We now run `H; while bypassRound; H; measure on 0`. The one-bit counter gave the first and third exits the same label. Here we measure the tag after every call. An exit on the first call therefore stays separate from every later exit.

The first Hadamard sends amplitude $1/\sqrt{2}$ down each path. On the first call the bypass exits, and so does $3/5$ of the lower path's amplitude. Both carry the same **`exit`** tag. At output 1 the second Hadamard takes their difference, giving amplitude $(1 - 3/5)/2 = 1/5$. The probability of exiting on this call and reading 1 is $1/25$.



Only the lower path can give **`back`**. Its returning amplitude is $(4/5)/\sqrt{2}$, so this outcome has probability $8/25$. Each further call returns again with probability $(3/5)^2$, so the loop eventually exits with probability one. Whenever it exits, its data is $|1\rangle$, and the second Hadamard gives 1 half the time. The first tag measurement distinguishes the two ways of reading 1. We therefore add their probabilities:

$$\underbrace{\frac{1}{25}}_{\text{exit on the first call}} + \underbrace{\frac{8}{25}\cdot\frac{1}{2}}_{\text{return on the first call}} = \frac{1}{5}.$$

A short pulse gives the same result in the ring. Only the light that goes straight past the ring arrives together with the bypass (Appendix H).

What one call measures. In `while f`, the body supplies the return type c, as in `trace f`. After the call we copy its tag to a fresh bit. If the exit carries x and the return carries y, the two clauses for the copy are

$$T : b + c \rightarrowtail (b + c) \otimes \mathbf{Qubit}, \qquad \mathrm{exit}\, x \mapsto \langle \mathrm{exit}\, x, 0\rangle, \quad \mathrm{back}\, y \mapsto \langle \mathrm{back}\, y, 1\rangle.$$

We then hide the bit, giving $\mathsf{hide}(\mathrm{arr}\,\beta; T)$. The bit distinguishes exit from return but carries no information about the payload within either tag. This is why both paths could still interfere on the first exit of `bypassRound`. Figure 8(a) shows the copy on the tag wire. Appendix C derives this body-first loop from the test-first loop of Ying and Feng when the external types agree [26, § 3].

In Figure 8(b) the return wire now carries a density block. We start with $(\rho, 0)$ because the caller supplies the data and nothing has returned yet. More generally, if ρ enters from outside and σ

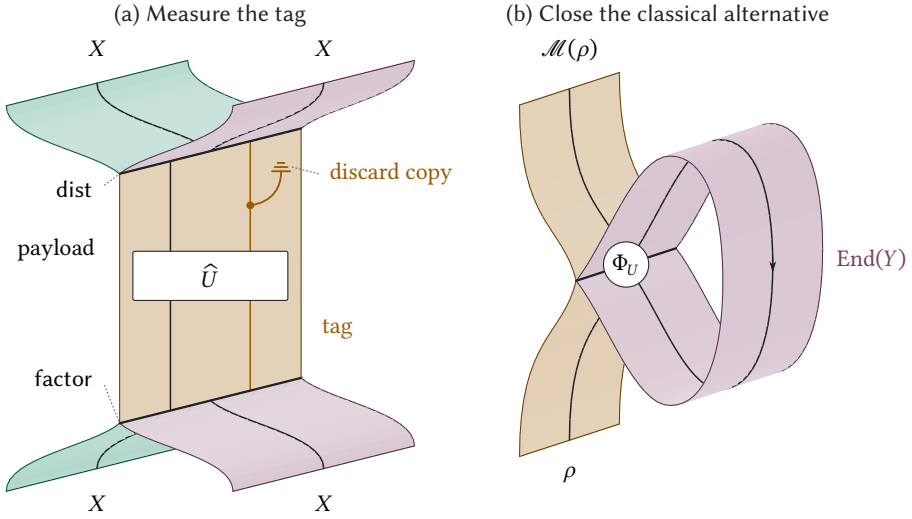


Fig. 8. Read upwards. At left, when $a = b = c$, factor exposes the tag as a separate wire. The box $\widehat{U}$ is the body in these coordinates. We discard a copy of the tag before dist restores the two ports (Appendix A.5). At right, only the return block of Φ_U goes around again. This construction also works when the payload types differ.

enters on the return wire, the measured call gives two output blocks,

$$\Phi_U(\rho, \sigma) = \big(\underbrace{A\rho A^\dagger + B\sigma B^\dagger}_{\text{exit}}, \underbrace{C\rho C^\dagger + D\sigma D^\dagger}_{\text{return}}\big).$$

The inputs are positive operators with total trace at most one. The trace of each output block is the probability on that branch. The first call emits $A\rho A^\dagger$ and returns $C\rho C^\dagger$. We feed only the return block into the second call, so the second exit is $(BC)\rho(BC)^\dagger$. Here A and BC are the first two exit maps from Section 3.

Adding the measured exits. An exit after k returns has given **back** on the first k calls and **exit** on the next. For any later exit, that same call gave **back**. Measuring the tag therefore separates every pair of different exit counts. For instance, the first two exits of `guardRound` have amplitudes $3/5$ and $-16/25$. In the coherent loop they add to $-1/25$ so far. In the measured loop the probability of having exited by the second call is $(3/5)^2 + (-16/25)^2$. Table 1 follows the next three calls.

After exactly k returns the exit map is x_k, so on input ρ that exit contributes $x_k \rho x_k^\dagger$. We keep that weight when we add the exits. Equivalently, we can keep the exact count from the preceding subsection and hide it at the end:

$$\mathcal{M}(\rho) = \sum_{k \geq 0} x_k \rho x_k^\dagger = \mathrm{tr}_{H_{\mathrm{rec}}}(R\rho R^\dagger), \qquad \mathcal{M} = [\![\mathsf{while}_c\, \beta]\!]. \tag{4.2}$$

The partial trace drops each term with two different count labels. Closing the return block of Φ_U gives this same sum. It is the trace $\mathrm{Tr}_{\mathrm{cl}}^{\mathrm{End}(Y)}(\Phi_U)$ for the classical sum in the category of completely positive trace-nonincreasing maps on finite C^*-algebras [3, Prop. 3.2.17]. For a finite unitary body, Theorem B.1 gives $\sum_k x_k^\dagger x_k = I$. For a normalized input the exit probabilities therefore sum to one. The measured run halts with probability one even when there is no last possible call.

Now consider the same finite body $\beta : a + c \leftrightarrow b + c$ in all three loops. Each uses the same exit maps x_k. In `trace` all the exits contribute to one sum. With a counter of $K \geq 1$ values, exits add when they carry the same count modulo K. In `while` we measure the tag on each call and add the resulting density matrices. For `bypassRound` on $|+\rangle$, the last two columns show what each loop returns and how often the caller reads 1 after the second Hadamard. Here $t = 3/5$ and $K = 2$. The caller hides the counter bit before applying the Hadamard.

Loop and type	*General output*	`bypassRound`	$\Pr(1)$
$\mathsf{trace}_c\ \beta : a \leftrightarrow b$	$Lu = \sum_{k\geq 0} x_k u$	$\lvert-\rangle$	1
$\mathsf{count}_K\ \beta : a \rightarrowtail [K] \otimes b$	$R_K u = \sum_{k\geq 0} \lvert k \bmod K\rangle \otimes x_k u$	$\dfrac{\lvert0\rangle \otimes \lvert0\rangle - \lvert1\rangle \otimes \lvert1\rangle}{\sqrt{2}}$	1/2
$\mathsf{while}_c\ \beta : a \rightsquigarrow b$	$\mathcal{M}(\rho) = \sum_{k\geq 0} x_k \rho x_k^\dagger$	$\frac{1}{2}\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}$	1/5

5 When can a caller forget the difference?

Suppose the caller needs only the data. Measuring the tag on each call gives the same result as keeping the exact count and hiding it at the end. In either case we add the contributions from different counts as density matrices. Closing the wire adds their amplitudes instead. We want to know when either choice gives the caller the same data on every input.

In `guardRound`, every exit carries `unit`. The first two exits multiply it by $3/5$ and $-16/25$, and each later exit picks up another factor of $3/5$. Their sum is $3/5 - (16/25)/(1 - 3/5) = -1$. The caller therefore receives `unit` with certainty whether we close the wire or measure the tag. Adding the bypass changes what an exit does to the input. The first exit carries amplitude from both paths, but every later exit carries only the part that went through the rotation. A return therefore singles out the rotation path.

For a general body, write $x_k u$ for the output after exactly k returns on input u. The coherent loop gives $Lu = \sum_k x_k u$. Suppose every x_k applies this same gate to the data, up to a scalar b_k that depends only on the count. Keeping the exact count then gives

$$Ru = \sum_{k\geq 0} |k\rangle \otimes x_k u = \left(\sum_{k\geq 0} b_k |k\rangle\right) \otimes Lu.$$

The count may vary from run to run, but its state is fixed across inputs. Hiding the count therefore leaves the caller with the coherent output Lu. We call this agreement between the measured and coherent loops *free erasure*.

Theorem 5.1 (Agreement on the returned data). *Let $\beta : a + c \leftrightarrow b + c$ be a finite iso with nonempty external input type a. Let $m = |\mathcal{B}(c)|$ be the number of basis values on the return wire. The following are equivalent.*

(1) The loops agree on the returned data, $\mathcal{M}(\rho) = L\rho L^\dagger$ for every input state ρ.
(2) One unit count vector $\xi \in \ell^2(\mathbb{N})$ satisfies $Ru = \xi \otimes Lu$ for every input u.
(3) One of $x_0, \ldots, x_m$ is nonzero and every map in this list is a scalar multiple of it.

In this case $x_k = b_k L$ for every k, where $\xi = \sum_k b_k |k\rangle$, $\sum_k b_k = 1$ and $\sum_k |b_k|^2 = 1$.

To see why agreement forces this factorization, give the coherent loop a unit input u. A measurement along any direction orthogonal to Lu has probability zero. In the measured loop each exit contributes its own nonnegative probability to that same outcome. For their sum to be zero, each $x_k u$ must lie along Lu. If an exit multiplied two basis outputs by different scalars, it would change their relative amplitudes on a superposition of those inputs. Thus at each count one scalar works for every input. For finitely many exits this also follows from the Kraus criterion [25, Cor. 2.23]. Appendix D gives the proof for the full sequence.

To check all the exits, we need only the four maps of the body. Write A for the immediate exit and C for the part of the first call that enters the return wire. On that wire B leads out and D goes around again. The exits in order are therefore

$$x_0 = A, \qquad x_1 = BC, \qquad x_2 = BDC, \qquad x_{j+1} = BD^jC \quad (j \geq 0).$$

If there are $m > 0$ return values, D is an m-by-m matrix. By Cayley–Hamilton, every power from D^m onward is a linear combination of $I, D, \ldots, D^{m-1}$. Multiplying on the left by B and on the right by C expresses every later exit as a linear combination of $x_1, \ldots, x_m$. When $m = 0$ there is only A.

We can therefore stop at x_m, even when there is no last round. At least one of these maps is nonzero, since their span contains the unitary L. We choose a nonzero entry in one map. For each other map, we divide its corresponding entry by the chosen one and check whether this ratio works for every entry. With decidable equality of exact coefficients, this gives a finite test.

A counter with $m > 0$ values can miss a difference that this test detects. An immediate exit and an exit after m returns both leave the counter at zero, so their amplitudes add. Measuring the tag distinguishes them on the first call.

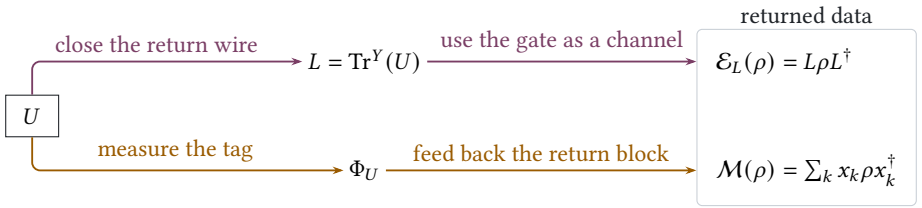


Fig. 9. Closing $U : X \oplus Y \to X' \oplus Y$ in Unitary gives $L : X \to X'$. Measuring the tag gives $\Phi_U : \mathrm{End}(X) \oplus \mathrm{End}(Y) \to \mathrm{End}(X') \oplus \mathrm{End}(Y)$, whose return we close by the classical-sum trace. Both outputs are channels in CPTP.

Replacing the loop in a program. When $x_k = b_k L$, the upper path in Figure 9 adds the exits to give $(\sum_k b_k)L = L$. On the lower path, exit k contributes $|b_k|^2 L\rho L^\dagger$. These weights sum to one, so the caller receives $L\rho L^\dagger$ along either path.

A program with a hole for this channel is a *channel context*. Equal channels give the same output probabilities even if the caller keeps a reference entangled with the input or runs the loop on one branch of a measurement. If they differ, some input and an output measurement distinguish them. A coherent bypass belongs to the body we test, before we give its channel to the caller [3, § 3.3.3].

Corollary 5.2 (Replacement in a caller). *Let β be a finite iso with m basis values on its return wire. Hiding the exact count always gives the measured loop. On a nonempty external input type, the measured loop may replace the coherent loop's data channel in every channel context if and only if $x_0, \ldots, x_m$ span a line. In that case, hiding the counter in $\mathrm{count}_K\ \beta$ gives the same channel for every $K \geq 1$. On an empty external input type all these channels agree.*

6 A count without a counter

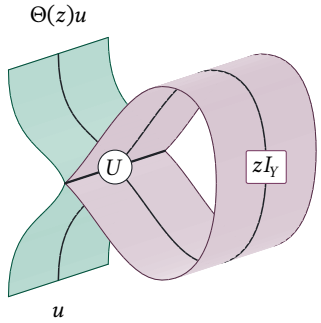
The agreement test concerns channels on returned data. We now keep the coherent iso before embedding it as a channel, so we can put it beside the reference used in Section 3. Changing the phase on its return wire will reveal how its response depends on the rounds.

A completed gate alone does not determine a count distribution: an immediate phase -1 and `guardRound` both give -1, but only `guardRound` has delayed exits. We distinguish them by inserting a phase $z = e^{i\varphi}$ on the return port and varying it. The new completed gate $\Theta(z)$ leaves an immediate

exit unchanged and multiplies exit k by z^k, one factor for each return. This *transfer function* therefore generates the original exits. Summing the marked exits will use an inverse on the returning space, so we first separate the vectors that never emit. Define

$$Y_0 = \{y \in Y \mid BD^j y = 0 \text{ for every } j \geq 0\}, \qquad Y_1 = Y_0^{\perp}.$$

By definition, $D(Y_0) \subseteq Y_0$ and $B|_{Y_0} = 0$. Unitarity gives $\|Dy\|^2 + \|By\|^2 = \|y\|^2$, so D preserves norm on Y_0. In finite dimension this restriction is onto. Unitarity then maps the orthogonal complements of these internal components onto one another, giving $C(X) \subseteq Y_1$ and $D(Y_1) \subseteq Y_1$. Put $B_1 = B|_{Y_1}$ and $D_1 = D|_{Y_1}$, regarding C as $X \to Y_1$. The geometric decay from Section 3 makes $I - zD_1$ invertible for $|z| \leq 1$, even when $I - zD$ on all of Y is singular.



	$j=0$	$j=1$	$j=2$
$k=0$	T_{00}	$z^{-1}T_{01}$	$z^{-2}T_{02}$
$k=1$	zT_{10}	T_{11}	$z^{-1}T_{12}$
$k=2$	z^2T_{20}	zT_{21}	T_{22}

$$T_{kj} = x_k \rho x_j^{\dagger}$$

$$\Theta_{\mathrm{rot}}(z) = \frac{t - z}{1 - tz},$$

$$\Theta_{\mathrm{bypass}}(z) = \mathrm{diag}(1, \Theta_{\mathrm{rot}}(z)).$$

$$\Theta(z) = \sum_{k \geq 0} z^k x_k = A + zB_1(I - zD_1)^{-1}C, \qquad L = \Theta(1).$$

Fig. 10. Each return adds a factor z at left. In the channel at right, exit pair k, j carries $z^{k-j}T_{kj}$. Averaging over the circle retains only the orange diagonal. The same cancellation applies to later exits. The displayed families Θ_{rot} and Θ_{bypass} belong to guardRound and bypassRound.

For the data channel, the term $x_k \rho x_j^{\dagger}$ carries phase $e^{i(k-j)\varphi}$, as the right panel of Figure 10 shows. Averaging over the circle keeps only $k = j$, so

$$\mathcal{M}(\rho) = \frac{1}{2\pi} \int_0^{2\pi} \Theta(e^{i\varphi})\, \rho\, \Theta(e^{i\varphi})^{\dagger}\, d\varphi. \tag{6.1}$$

Thus the measured loop is a uniform mixture of coherent loops differing only in their return phase. Geometric decay justifies these Fourier manipulations.

Differentiation can recover a count because a single exit at count k has response $e^{ik\varphi}L$. Its derivative is ik times that response. Undoing the gate and multiplying by $-i$ gives kI. With several exits there are also cross terms at a fixed phase. The unitary transfer identity and the phase average from monitored recurrence [5, Eq. (4.10)] [6, § 3, Eq. (8)] relate this coherent observable to the measured mean.

Theorem 6.1 (The measured mean and the phase derivative). *Let $F_z = (I - zD_1)^{-1}C$ and $N = \sum_{k \geq 0} k\, x_k^{\dagger} x_k$. For $z = e^{i\varphi}$, the positive semidefinite relative phase derivative and its average satisfy*

$$Q(z) = -i\Theta(z)^{\dagger} \frac{d}{d\varphi} \Theta(e^{i\varphi}) = F_z^{\dagger} F_z,$$

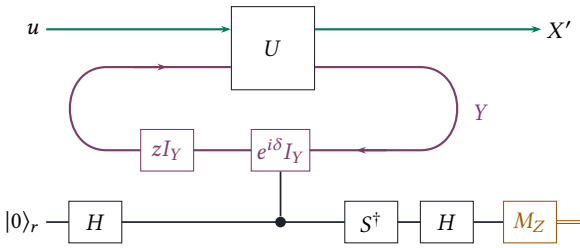
$$\frac{1}{2\pi} \int_0^{2\pi} Q(e^{i\varphi})\, d\varphi = N, \qquad \mathrm{tr}\, N = \dim Y_1.$$

For unit $u \in X$, $\langle u, Nu \rangle$ is the measured mean number of returns.

Appendix E differentiates U's inner-product identity and averages over phase. With $CC^\dagger = I_{Y_1} - D_1 D_1^\dagger$ from unitarity, cyclicity and decay give the telescope

$$\operatorname{tr} N = \sum_{j\ge 0} \operatorname{tr}\big(D_1^j C C^\dagger (D_1^\dagger)^j\big) = \lim_{n\to\infty} \operatorname{tr}\big(I_{Y_1} - D_1^n (D_1^\dagger)^n\big) = \dim Y_1.$$

For a unit input u, prepare a reference bit in $|+\rangle$ to control only the choice of return phase z or $ze^{i\delta}$. The $S^\dagger; H$ readout in Figure 11 measures the outputs' imaginary overlap, with inner products linear in the second argument [7, p. 1, Eqs. (1)–(3)]. Appendix I gives its pure iso construction. Since $Q(z) = F_z^\dagger F_z$, the first-order response gives the squared norm of the returning amplitude before the phase gate. Several rounds contribute to this wire, so its circulating weight can exceed one.



$U : X \oplus Y \to X' \oplus Y$

guardRound, $u = 1$

z	F_z	$Q(z)$
1	$s/(1-t) = 2$	4
-1	$s/(1+t) = 1/2$	$1/4$

$S^\dagger = \operatorname{diag}(1, -i)$

$$\Pr(0) - \Pr(1) = \operatorname{Im}\langle \Theta(z)u, \Theta(ze^{i\delta})u\rangle = \delta\langle u, Q(z)u\rangle + O(\delta^2).$$

Fig. 11. The reference chooses return phase z or $ze^{i\delta}$ and accompanies both summands unchanged through U. Its final measurement reads the relative derivative. At right, $t = 3/5$, $s = 4/5$, and F_z is the scalar return amplitude before the phase gate.

For guardRound, the measured mean is $\sum_{k\ge1} k s^4 t^{2(k-1)} = \sum_{j\ge0} s^2 t^{2j} = s^2/(1-t^2) = 1$. The second sum adds survival probabilities. The table beside the circuit instead squares the summed return amplitudes, giving coherent weights 4 and 1/4. Averaging $Q(z)$ over all phases recovers the measured mean.

The bypass in bypassRound makes no return, giving $N_{\text{bypass}} = \operatorname{diag}(0, 1)$. Its measured mean on $|+\rangle$ is therefore $1/2$, the equal-weight average of bypass count zero and scalar-loop mean one.

Putting return phase $z = e^{i\varphi}$ into the original Hadamard caller gives these observable probabilities:

$$\Pr(b = 1) = |1 - \Theta_{\text{rot}}(z)|^2/4$$

φ	0	$\pi/2$	π
$\Theta_{\text{rot}}(e^{i\varphi})$	-1	$(15-8i)/17$	1
$\Pr(b=1)$	1	$1/17$	0

Averaging this test over the phase circle gives the measured probability $1/5$ by (6.1).

On a nonempty external type, agreement gives $\Theta(z) = f(z)L$ with $|f(z)| = 1$ on the circle, so every return phase gives the same data channel. Conversely, if every phase channel equals $\mathcal{E}_L$, their average is both $\mathcal{E}_L$ and $\mathcal{M}$ by (6.1). Thus the finite test also decides whether changing the return phase can affect the data.

Appendix H realizes the scalar response with an ideal lossless ring coupler [2, 21].

7 The three readings in the Π family

Figure 12 places the three readings in the Π family. A backwards wire is supplied by the standard Int construction: its objects are pairs (P^+, P^-), and a map to (Q^+, Q^-) is a unitary $P^+ \oplus Q^- \to Q^+ \oplus P^-$. The fully faithful embedding $N(X) = (X, 0)$ preserves the sum trace [14, 18]. Appendix G draws the typed components of this bending.

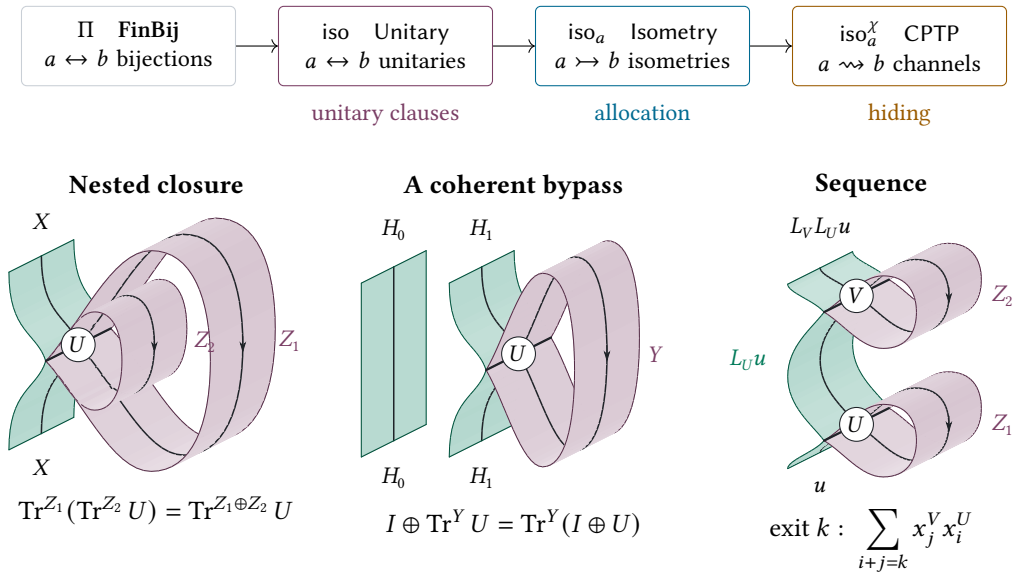


Fig. 12. Finite source extensions and coherent trace laws for endomorphisms [4, 11, 14]. Allocation gives $R[\text{Unitary}] \simeq \text{Isometry}$ and hiding gives $L[\text{Isometry}] \simeq \text{CPTP}$. Here $L = \sum_k x_k$ for each body, with returns kept in that body and canonical reassociations.

AI Use. The author used ChatGPT Codex (GPT-6.1 Sol) to assist with paper formatting, generating TikZ for figures, exploratory coding, literature searches, and brainstorming of analytical approaches. The author verified all results, revised the manuscript as needed, and takes full responsibility for its content.

Data-Availability Statement. The finite iso interpreter uses floating-point arithmetic and cyclic recording. We intend to submit it and its examples for Artifact Evaluation.

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A Typing and interpreting the clauses

A.1 Syntax

$$
\begin{aligned}
a, b &::= 0 \mid 1 \mid a + b \mid a \otimes b \\
v &::= * \mid x \mid \mathsf{left}\ v \mid \mathsf{right}\ v \mid \langle v_1, v_2 \rangle \mid \mathsf{exit}\ v \mid \mathsf{back}\ v \\
p, q &::= * \mid x \mid \langle p, q \rangle \\
e &::= v \mid \textstyle\sum_i \alpha_i e_i \mid \mathsf{let}\ p = \omega\ q\ \mathsf{in}\ e \\
\omega &::= f \mid \{v_i \leftrightarrow e_i\}_{i \in I} \mid \omega\,;\,\omega \mid \omega \oplus \omega \mid \omega \otimes \omega \mid \omega^\dagger \mid \mathsf{trace}_c\ \omega \\
t &::= e \mid \omega\ t \\
\mathit{decl} &::= f :: a \leftrightarrow b\ ;\ f = \omega \mid \mathsf{round}\ f : a + c = \{v_i \leftrightarrow e_i\}_{i \in I}
\end{aligned}
$$

Fig. 13. Finite iso definitions. A name f refers to an earlier closed definition. The coefficient α_i is a complex scalar. A signature uses ::, while the **round** abbreviation uses :. The subscript in trace_c identifies the return type.

The surface patterns exit v and back v elaborate recursively to left v and right v. This rewrite preserves payloads and nesting, and is performed before matching or typing by the core rules below. The remaining executable abbreviations are those of Section 2: **Unit**, `unit`, **Qubit** and its basis values, bracketed scalar combinations, combinator composition and on application. Listings typeset the source token <-> as ↔ and pair delimiters <, > as ⟨, ⟩.

A.2 Statics

Admission, elaboration and evaluation are defined together by induction over the ordered declarations and proper subprograms. In particular, evaluating a local call to fill a table column uses a proper subprogram or an already checked earlier definition. The grammar in Figure 13 describes definitions in an ordered environment Σ. A binding $f = \omega : a \leftrightarrow b$ is introduced after checking ω against earlier bindings. Write Σ ok when these declarations form a well-typed environment. Types are the finite trees of Figure 13. Both 0 and 1 are types, and $a + b$ and $a \otimes b$ are types whenever a and b are. Names may be chosen fresh by renaming. The rules for introducing them are

$$
\frac{}{\varnothing\ \mathsf{ok}} \qquad \frac{\Sigma\ \mathsf{ok} \quad f \notin \mathrm{dom}(\Sigma) \quad \Sigma \vdash \omega : a \leftrightarrow b}{\Sigma, (f = \omega : a \leftrightarrow b)\ \mathsf{ok}}.
$$

The declaration

$$
\mathsf{round}\ f : a + c = \{v_i \leftrightarrow e_i\}_{i \in I}
$$

introduces the ordinary binding $f = \{v_i \leftrightarrow e_i\}_{i \in I} : (a + c) \leftrightarrow (a + c)$ after the literal clauses have passed the same check. The name denotes the open round. Closing it requires an explicit operation.

Writing $\Sigma \vdash d \Rightarrow \Sigma'$ for declaration elaboration, its rule is

$$\frac{\Sigma \text{ ok} \quad f \notin \mathrm{dom}(\Sigma) \quad \Sigma \vdash \mathcal{T} : (a + c) \leftrightarrow (a + c)}{\Sigma \vdash \textbf{round}\ f : a + c = \mathcal{T} \Rightarrow \Sigma, (f = \mathcal{T} : (a + c) \leftrightarrow (a + c))}.$$

An ordinary iso signature still permits distinct external types. The binding $f = \omega : a \leftrightarrow b$ represents the surface declaration $f :: a \leftrightarrow b\ ;\ f = \omega$; listings may separate these parts by a newline. Write Δ for a finite set of distinct typed value variables. In Δ_1, Δ_2 the two domains must be disjoint. Each judgment below uses this convention, including inside a clause with an empty set of variables. The empty table presents the unique map on 0. Whenever we use a unit input or a nonzero operator line we assume the corresponding external type is inhabited.

Patterns and expressions. The finite basis is defined recursively:

$$\mathcal{B}(0) = \varnothing, \qquad \mathcal{B}(1) = \{*\},$$
$$\mathcal{B}(a + b) = \mathrm{left}\,\mathcal{B}(a) \sqcup \mathrm{right}\,\mathcal{B}(b), \quad \mathcal{B}(a \otimes b) = \mathcal{B}(a) \times \mathcal{B}(b).$$

An environment for Δ assigns each variable a basis value of its type. A variable matches any such value; a constructor matches its own tag, and a pair matches its two components. Write $\Delta \vdash v : a$ when a basis pattern is well typed. The rules are

$$\varnothing \vdash * : 1 \qquad x : a \vdash x : a \qquad \frac{\Delta \vdash v : a}{\Delta \vdash \mathrm{left}\ v : a + b} \qquad \frac{\Delta \vdash v : b}{\Delta \vdash \mathrm{right}\ v : a + b}$$

$$\frac{\Delta_1 \vdash v : a \qquad \Delta_2 \vdash w : b}{\Delta_1, \Delta_2 \vdash \langle v, w \rangle : a \otimes b}.$$

The basis assignments for a context Δ form $\mathcal{B}(\Delta) = \prod_{(x:a) \in \Delta} \mathcal{B}(a)$. The empty product contains the unique empty assignment. If any variable has an empty basis, the product is empty. Write $\mathcal{H}(\Delta) = \mathbb{C}^{\mathcal{B}(\Delta)}$. A product pattern p with $\Delta \vdash p : b$ gives a bijection $\gamma \mapsto p[\gamma]$ from $\mathcal{B}(\Delta)$ to $\mathcal{B}(b)$. For unit it maps the unique empty assignment to $*$. For a variable it is the identity. For a pair, disjoint contexts let us combine the two component assignments uniquely. Induction on these three rules proves the bijection, including empty bases. Matching p is its inverse.

The unit, variable and pair rules also type the product patterns p, q in Figure 13. In the argument position q these rules assemble already bound values. In the result position p they bind fresh variables of the indicated types. A single variable in either position may stand for a value of any type. For expressions we write $\Sigma; \Delta \vdash e : a$ and use

$$\frac{\Delta \vdash v : a}{\Sigma; \Delta \vdash v : a} \qquad \frac{\Sigma; \Delta \vdash e_i : a\ (i \in I) \qquad \alpha_i \in \mathbb{C}}{\Sigma; \Delta \vdash \sum_{i \in I} \alpha_i e_i : a}$$

$$\frac{\Delta_1 \vdash q : a \qquad \Sigma \vdash \omega : a \leftrightarrow b \qquad \Delta_p \vdash p : b \qquad \Sigma; \Delta_2, \Delta_p \vdash e : d}{\Sigma; \Delta_1, \Delta_2 \vdash \mathsf{let}\ p = \omega\ q\ \mathsf{in}\ e : d}.$$

In the last rule the variables in p are fresh and pairwise distinct. A product pattern of type b binds every basis output, by the bijection above. The sum rule permits an empty index set in any stated context Δ; its denotation is the zero linear map from $\mathcal{H}(\Delta)$. Such an expression cannot supply an entire column of a nonempty unitary table. A finite first-order term is a closed expression, or an application of a closed iso to a term of its domain. Writing $\Sigma \vdash t : a$ for a closed term, its rules are

$$\frac{\Sigma; \varnothing \vdash e : a}{\Sigma \vdash e : a} \qquad \frac{\Sigma \vdash t : a \qquad \Sigma \vdash \omega : a \leftrightarrow b}{\Sigma \vdash \omega\, t : b}.$$

This suffices for feeding a superposition of ground values to a program.

For a matrix M with input columns s and output rows t, row s of $M^\dagger$ is the conjugate of column s of M. Consequently its Gram matrix has entries

$$(M^\dagger M)_{ss'} = \sum_{t \in \mathcal{B}(b)} \overline{M_{ts}} M_{ts'}.$$

Each product pairs amplitudes in the same output row. At $s = s'$ it contributes to the squared norm of one column; at $s \neq s'$ it contributes to the inner product between two columns.

Let M have one column for each $s \in \mathcal{B}(a)$, filled with the coefficients of the expression in the uniquely matching clause under its ground substitution. The following rule imposes a finite extensional check. Its orthogonality premise concerns fully expanded outputs, so it need not infer linear independence from surface syntax.

$$\frac{\begin{array}{c}\Delta_i \vdash v_i : a \quad \Sigma; \Delta_i \vdash e_i : b \quad (i \in I) \\ \big(v_i[\mathcal{B}(\Delta_i)]\big)_{i \in I} \text{ disjointly covers } \mathcal{B}(a) \qquad |\mathcal{B}(a)| = |\mathcal{B}(b)| \quad M^\dagger M = I\end{array}}{\Sigma \vdash \{v_i \leftrightarrow e_i\}_{i \in I} : a \leftrightarrow b}$$

Here $v_i[\mathcal{B}(\Delta_i)]$ is the set of all ground instances of pattern v_i. The coverage premise requires these sets to be disjoint and their union to be $\mathcal{B}(a)$. An empty instance set is permitted and contributes no column. This rule is exact over a specified coefficient domain with exact equality and arithmetic, for example rational complex numbers. With mathematical complex parameters it is a well-formedness predicate; exact equality of arbitrary descriptions of complex numbers is not assumed decidable. The finite semantics does not make floating-point tolerance an orthogonality proof.

Reversible combinators. For $M \in \mathbb{C}^{\mathcal{B}(b) \times \mathcal{B}(a)}$ the atom rule is

$$\frac{|\mathcal{B}(a)| = |\mathcal{B}(b)| \qquad M^\dagger M = I}{[M] : a \leftrightarrow b}.$$

Write u, v for reversible programs and omit the common Σ in the following rules. A bound name has the type at its declaration. For finite types we have the structural isos of Π as abbreviations for the corresponding permutation clauses, including identity, symmetries and distributivity. The reversible rules are

$$\frac{f : a \leftrightarrow b \in \Sigma}{f : a \leftrightarrow b} \qquad \frac{u : a \leftrightarrow b \qquad v : b \leftrightarrow d}{u \,;\, v : a \leftrightarrow d} \qquad \frac{u : a \leftrightarrow b}{u^\dagger : b \leftrightarrow a}$$

$$\frac{u : a \leftrightarrow b \qquad v : d \leftrightarrow f}{u \oplus v : a + d \leftrightarrow b + f} \qquad \frac{u : a \leftrightarrow b \qquad v : d \leftrightarrow f}{u \otimes v : a \otimes d \leftrightarrow b \otimes f} \qquad \frac{\beta : a + c \leftrightarrow b + c}{\mathrm{trace}_c\, \beta : a \leftrightarrow b}.$$

A.3 Dynamics

Evaluating an expression. Let $\gamma \in \mathcal{B}(\Delta)$ assign basis values to the variables in a linear context Δ, and write $\gamma \vdash e \Downarrow \psi$ when evaluation returns the finite output vector ψ. For a value pattern v, $v[\gamma]$ is its ground instance. Values and sums evaluate by

$$\frac{}{\gamma \vdash v \Downarrow |v[\gamma]\rangle} \qquad \frac{\gamma \vdash e_i \Downarrow \psi_i \ (i \in I)}{\gamma \vdash \sum_i \alpha_i e_i \Downarrow \sum_i \alpha_i \psi_i}.$$

For let $p = \omega q$ in e, split the input assignment as $\gamma_1 \uplus \gamma_2$ according to the typing rule: q uses precisely γ_1, and e retains γ_2. Write $U|q[\gamma_1]\rangle = \sum_t a_t |t\rangle$, where $\omega \Downarrow U$. Matching the product pattern p with

t gives the fresh assignment γ_t. The application rule is

$$\frac{\omega \Downarrow U \quad \gamma_2 \uplus \gamma_t \vdash e \Downarrow \psi_t \quad (t \in \mathcal{B}(b))}{\gamma_1 \uplus \gamma_2 \vdash \mathsf{let}\ p = \omega q\ \mathsf{in}\ e \Downarrow \sum_t a_t \psi_t}.$$

Here b is the output type of ω. Its admitted matrix supplies the coefficients a_t, and total matching of p supplies each γ_t. The same binding is used independently in the alternatives of a sum; it is split only where the linear typing rule splits it. For closed terms, $t \Downarrow \psi$ uses the empty variable assignment, with

$$\frac{\varnothing \vdash e \Downarrow \psi}{e \Downarrow \psi} \qquad \frac{\omega \Downarrow U \qquad t \Downarrow \psi}{\omega t \Downarrow U\psi}.$$

The fixed iso environment is omitted from these evaluation judgments. Named matrices are computed from earlier definitions; the reversible combinators act by their matrix operations, and a trace uses Proposition B.2. Thus expression evaluation is finite even when its matrix contains feedback with infinitely many nonzero exits. Induction on an expression's typing derivation shows that evaluation gives its linear interpretation: the value rule substitutes basis values, the sum rule is linearity, and the application rule composes the admitted iso with the continuation's linear map. These rules do not select a measurement outcome or normalize individual amplitude components. More explicitly, the local application's column at (γ_1, γ_2) is

$$E_{\mathsf{let}}(s; \gamma_1, \gamma_2) = \sum_{t \in \mathcal{B}(b)} U_{t,q[\gamma_1]}\, E_e(s; \gamma_2, \gamma_t).$$

Here $s \in \mathcal{B}(d)$ indexes the continuation output, and $t \in \mathcal{B}(b)$ indexes the intermediate output of the local iso. This is ordinary matrix multiplication: reassemble q's variables, apply U while carrying γ_2, match its output to the fresh product pattern, and apply the continuation matrix. The two pattern maps are the bijections just proved. Thus the rule acts linearly on arbitrary superpositions of assignments, including entangled ones. It neither duplicates an assignment nor measures the local result.

Elaboration and matrix evaluation. The matrix programs have grammar

$$\kappa ::= [M] \mid \kappa; \kappa \mid \kappa \oplus \kappa \mid \kappa \otimes \kappa \mid \kappa^\dagger \mid \mathsf{trace}_c\, \kappa.$$

The combinators κ of Section 2 have ground checked tables as atoms. Write $\mathcal{E}_\Sigma(\omega)$ for elaboration. If $f = \omega_f$ is a declaration in Σ, let Σ_f be its earlier prefix, the environment in which ω_f was checked. With M the evaluated columns of a table $\mathcal{T}$, the definition is

$$\begin{aligned}
\mathcal{E}_\Sigma(\mathcal{T}) &= [M], & \mathcal{E}_\Sigma(f) &= \mathcal{E}_{\Sigma_f}(\omega_f),\\
\mathcal{E}_\Sigma(\omega; \nu) &= \mathcal{E}_\Sigma(\omega); \mathcal{E}_\Sigma(\nu), & \mathcal{E}_\Sigma(\omega \oplus \nu) &= \mathcal{E}_\Sigma(\omega) \oplus \mathcal{E}_\Sigma(\nu),\\
\mathcal{E}_\Sigma(\omega \otimes \nu) &= \mathcal{E}_\Sigma(\omega) \otimes \mathcal{E}_\Sigma(\nu), & \mathcal{E}_\Sigma(\omega^\dagger) &= \mathcal{E}_\Sigma(\omega)^\dagger,\\
\mathcal{E}_\Sigma(\mathsf{trace}_c\, \omega) &= \mathsf{trace}_c\, \mathcal{E}_\Sigma(\omega).
\end{aligned}$$

These equations recurse into proper subprograms and, on lookup, earlier definitions. Computing a table uses this same evaluation on its local iso subprograms. This is a well-founded mutual definition with expression evaluation. Each clause of the definition preserves its source and target types. The declaration rule for **round** binds this open combinator and does not insert a trace.

If $\kappa \Downarrow M$ and $\lambda \Downarrow N$, the ordinary matrix rules, used with their typed premises, are

$$\begin{aligned}
&[M] \Downarrow M, && (\kappa; \lambda) \Downarrow NM,\\
&(\kappa \oplus \lambda) \Downarrow M \oplus N, && (\kappa \otimes \lambda) \Downarrow M \otimes N, \quad \kappa^\dagger \Downarrow M^\dagger.
\end{aligned}$$

The feedback rule completes them:

$$\frac{\kappa \Downarrow U \quad U = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \quad F : X \to Y \quad (I_Y - D)F = C}{\mathsf{trace}_c\, \kappa \Downarrow A + BF}.$$

The premise solves a finite linear system. It neither inverts a possibly singular $I - D$ nor requires an execution to reach a last exit. Theorem B.1 gives a solution and proves that every solution gives the same output. Proposition B.2 computes that output by eliminating one return pattern at a time. Thus this relational evaluation has a result independent of the chosen wire solution, and exact elimination implements it. Write $\Sigma \vdash \omega \Downarrow M$ when its elaboration evaluates to M. Closed application then uses

$$\frac{\Sigma \vdash \omega \Downarrow M \qquad t \Downarrow \psi}{\omega t \Downarrow M\psi}.$$

For a well-typed expression and a basis assignment, induction on its syntax gives a unique output vector. In the local-binding case, the argument and result patterns give inverse assembly/matching bijections, and the finite coefficient sum is the matrix product displayed above. Expanding each clause therefore preserves all its columns. Induction through the combinators preserves evaluation and the linear denotation; the trace case uses the feedback theorem. Linearity then gives the same result on arbitrary superpositions, including assignments entangled with a retained factor.

The executable accepts unit and nested local patterns, empty types and tables, and the `dagger` combinator for adjoint. Its combinator spellings are ; for sequence, + for direct sum and $*$ for tensor. Native numerical evaluation uses floating-point coefficients and may refuse a coupled near-unit pivot; it is not the exact field-arithmetic procedure of Proposition B.2.

A.4 Denotation

A checked table denotes the linear extension of its columns. Names, sequence, dagger, direct sum and tensor denote lookup, composition in execution order, adjoint, direct sum and tensor; trace denotes the exit sum (3.2). The evaluation identities above give these same maps, with the trace case supplied by Theorem B.1.

Control of an endo-iso u is $I \oplus u$ under the distributive iso for $(1 + 1) \otimes a$. Thus controlling a global phase can change a relative phase. The types above deliberately do not provide this operation on an arbitrary channel.

A.5 Effect extensions

Syntax. Write u for a reversible iso, v for an isometry program and h for a channel program. The two upper grammars are

$$v ::= \mathsf{arr}\, u \mid \mathsf{lift}\, u \mid v\,;\, v \mid v \oplus v \mid v \otimes v \mid \mathsf{count}_K\, u,$$
$$h ::= \mathsf{arr}\, v \mid \mathsf{hide}\, v \mid h\,;\, h \mid h \otimes h \mid \mathsf{while}_c\, u.$$

Statics. The type of the intermediate result must match the type expected by the next program. At the isometry layer this gives the following rules for sequence, tensor and direct sum.

$$\frac{v : a \rightarrowtail b \qquad w : b \rightarrowtail d}{v\,;\, w : a \rightarrowtail d} \qquad \frac{v : a \rightarrowtail b \qquad w : c \rightarrowtail d}{v \otimes w : a \otimes c \rightarrowtail b \otimes d} \qquad \frac{v : a \rightarrowtail b \qquad w : c \rightarrowtail d}{v \oplus w : a + c \rightarrowtail b + d}.$$

Channels also compose in sequence and by tensor product.

$$\frac{h : a \rightsquigarrow b \qquad k : b \rightsquigarrow d}{h\,;\, k : a \rightsquigarrow d} \qquad \frac{h : a \rightsquigarrow b \qquad k : c \rightsquigarrow d}{h \otimes k : a \otimes c \rightsquigarrow b \otimes d}.$$

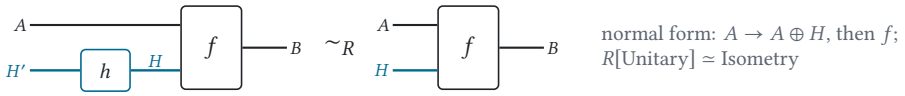
These operations have their usual linear meanings, with sequence written in execution order. An identity on each layer is the embedded identity iso. The remaining rules change the kind of map or close a round.

$$\frac{u : a \leftrightarrow b}{\mathsf{arr}\, u : a \rightarrowtail b} \qquad \frac{u : a + d \leftrightarrow b}{\mathsf{lift}\, u : a \rightarrowtail b} \qquad \frac{\beta : a + c \leftrightarrow b + c \qquad K \geq 1}{\mathsf{count}_K\, \beta : a \rightarrowtail [K] \otimes b}$$

$$\frac{v : a \rightarrowtail b}{\mathsf{arr}\, v : a \rightsquigarrow b} \qquad \frac{v : a \rightarrowtail b \otimes g \qquad |\mathcal{B}(g)| > 0}{\mathsf{hide}\, v : a \rightsquigarrow b} \qquad \frac{\beta : a + c \leftrightarrow b + c}{\mathsf{while}_c\, \beta : a \rightsquigarrow b}.$$

Denotation. An $R[\mathsf{Unitary}]$ representative $U : X \oplus H \to X'$ exposes $u \mapsto U(u, 0)$, identifying changes confined to the unused summand. An $L[\mathsf{Isometry}]$ representative $V : X \to X' \otimes G$ exposes $\rho \mapsto \mathrm{tr}_G(V \rho V^\dagger)$, identifying isometric relabelings of the hidden factor and the equivalence relation they generate [11, Prop. 7, Thm. 22].

(a) $R[\mathscr{C}]$: a heap enters beside the input, and only access to it counts



(b) $L[\mathscr{C}]$: garbage leaves beside the output, and its contents do not count

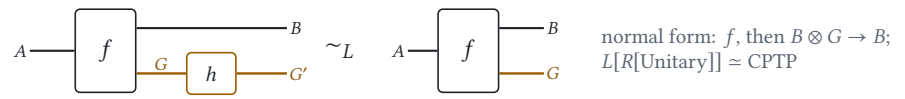


Fig. 14. Allocation exposes an input summand of a larger reversible map. Hiding exposes one output factor of an isometry. Preprocessing the unused summand or isometrically relabeling the hidden factor preserves the exposed map. Here $C = \mathsf{Unitary}$ in $R[C]$ and $C = \mathsf{Isometry}$ in $L[C]$.

For $0 \leq j < K$, the allocation permutation $p_j : a + ([K-1] \otimes a) \leftrightarrow [K] \otimes a$ maps left x to $\langle j, x \rangle$, taking $[0] = 0$. Then $\mathsf{alloc}_j = \mathsf{lift}\, p_j$. The basis copier for inhabited b uses heap $[|\mathcal{B}(b)| - 1] \otimes b$, with $|\mathcal{B}(b)|^2 - |\mathcal{B}(b)|$ basis values, completing the diagonal injection to a permutation. An embedded unitary is the same map viewed as an isometry. An embedded isometry V is the channel $\mathcal{E}_V$, distinct from the syntactic elaboration $\mathcal{E}_\Sigma$. For lift and hide the denotations are restriction and partial trace as in Section 4. For count_K, tensor $I_{[K]}$ with β, distribute the tensor over the two tags and append $I_{[K] \otimes b} \oplus (S_K \otimes I_c)$ in the resulting order. Trace the summand $[K] \otimes c$, then precompose with $u \mapsto |0\rangle \otimes u$. This allocation is a lift of a permutation that places the exposed basis at counter value zero. The four blocks of the enlarged iso are $I \otimes A, I \otimes B, S_K \otimes C, S_K \otimes D$. Its kth exit is $S_K^k \otimes x_k$, proving (4.1). To hide the counter in the displayed syntax, compose with $\mathsf{arr}\, \sigma_{[K],b}$ before hide. Native `count K beta` returns data before counter, so its following `discard` hides the counter. Appending `swap` instead gives our displayed order. In the executable, `discard` is the derived first projection, $I_a \otimes \mathsf{discard}_b; \mathsf{unit}_\otimes$, rather than the whole-register operation $\mathsf{discard}_b : b \rightsquigarrow 1$.

For while, allocate a fresh bit at 0 and apply the ordinary iso

$$\omega_{\mathrm{tag}} = \{ \quad \langle \mathsf{exit}\, x, 0\rangle \leftrightarrow \langle \mathsf{exit}\, x, 0\rangle, \quad \langle \mathsf{exit}\, x, 1\rangle \leftrightarrow \langle \mathsf{exit}\, x, 1\rangle, \quad \langle \mathsf{back}\, y, 0\rangle \leftrightarrow \langle \mathsf{back}\, y, 1\rangle, \quad \langle \mathsf{back}\, y, 1\rangle \leftrightarrow \langle \mathsf{back}\, y, 0\rangle \quad \} : (b + c) \otimes \mathbf{Qubit} \leftrightarrow (b + c) \otimes \mathbf{Qubit}.$$

The patterns partition the basis and merely permute its values, so the column test admits this iso. Its restriction to bit input 0 gives the isometry T. For output components $v \in [\![b]\!]$ and $y \in [\![c]\!]$, its linear action is

$$T(v, y) = (\iota_{\mathrm{ext}} v) \otimes |0\rangle + (\iota_{\mathrm{ret}} y) \otimes |1\rangle.$$

For the empty type, $\mathsf{discard}_0$ is obtained by hiding the unit factor of the unique isometry $0 \rightarrowtail 1 \otimes 1$, constructed by lift from $0 + 1 \leftrightarrow 1 \otimes 1$. Its action is the unique map on the zero operator space. Set $\mathsf{measure}_0$ to the embedded identity on 0. These supply the empty cases omitted from the inhabited-type formulas in the body.

Applying T after U and tracing out the copied bit gives the diagonal output blocks. Feedback now uses the classical sum. This construction works when the payload types differ. The notation while names this classical feedback operation on the translated body.

When $a = b = c$, let factor : $a + a \leftrightarrow a \otimes \mathbf{Qubit}$ and dist = factor†, and put $\widehat{\beta} = \mathsf{dist}; \beta; \mathsf{factor}$. Writing $E(\omega) = \mathsf{arr}(\mathsf{arr}\, \omega)$, the channel in Figure 8 is

$$E(\mathsf{factor}); E(\widehat{\beta}); (I_a \otimes \mathsf{measure}_{\mathbf{Qubit}}); E(\mathsf{dist}).$$

The first factor; dist cancel, leaving the original body followed by exposure, measurement and restoration of its tag.

Soundness. Induct on the finite syntax in a well-founded order that puts prior definitions before uses. The unique match and the substitution clause give a column at each closed input. The admission premise makes these columns an orthonormal basis. Sequential, sum, tensor and dagger operations preserve the stipulated map classes. Theorem B.1 proves the trace case. Restriction of a unitary is an isometry, and a unitary after counter allocation is an isometry. Embedding an isometry and taking a partial trace after an isometry both give completely positive trace-preserving maps. The positive sum in (4.2) is completely positive, since each finite partial sum is and the completely positive cone in finite dimension is closed; its trace preservation follows from $\sum_k x_k^\dagger x_k = I$. Thus every well-typed iso, isometry program and channel program denotes the indicated kind of map. A closed term $t : a$ instead denotes its evaluated vector in $\mathbb{C}^{\mathcal{B}(a)}$; its typing alone does not require that vector to have norm one. Conversely a given finite unitary V has as a definition the table $s \leftrightarrow \sum_t V_{ts} t$. Each column satisfies the typing rule since $V^\dagger V = I$, including the empty case.

B Convergence, probability and elimination

For a finite permutation, an external input follows a single path. It cannot revisit an internal basis value: unique predecessors would identify its original external input with an internal value. It therefore exits within $|\mathcal{B}(c)| + 1$ calls, while wholly internal cycles remain unreachable. The two external paths in Figure 15 combine to a swap, as in reversible flowchart semantics [10, Prop. 3.10][15, Prop. 5.3].

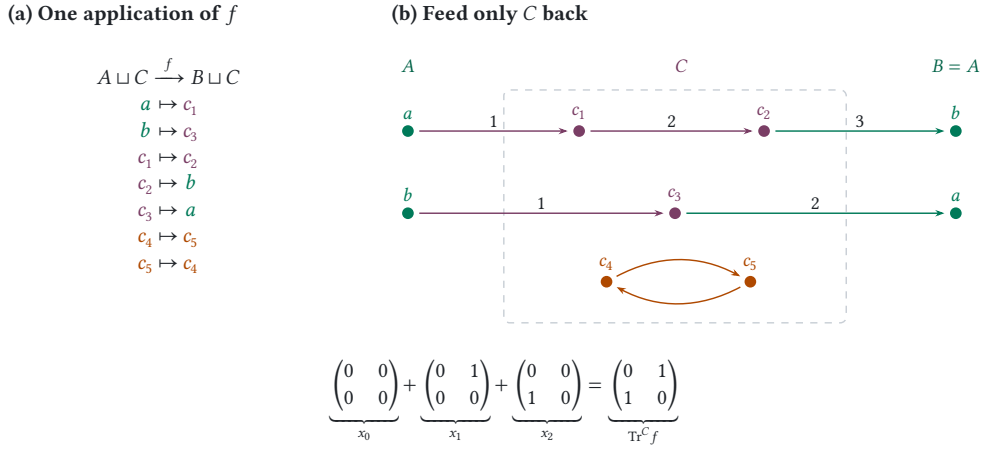
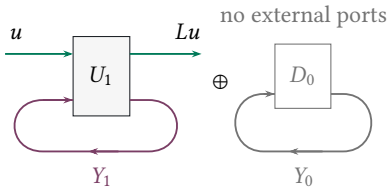


Fig. 15. A reversible loop with two external paths and an internal cycle. The two nonzero exit maps have disjoint domains and their sum is a swap.

Theorem B.1 (Finite feedback). *For a finite unitary $U : X \oplus Y \to X' \oplus Y$, the exits are absolutely summable in operator norm. Their sum $L : X \to X'$ is unitary, and $\sum_k x_k^\dagger x_k = I_X$. The wire equation $y = Cu + Dy$ has a solution for every u. Every solution gives the same output $Au + By = Lu$.*



$$\|u\|^2 = \|Au\|^2 + \|Cu\|^2,$$
$$\|Cu\|^2 = \|BCu\|^2 + \|DCu\|^2,$$
$$\|DCu\|^2 = \|BDCu\|^2 + \|D^2Cu\|^2.$$

Fig. 16. An internal basis change separates U_1 from the unreachable D_0. The three norm identities telescope because each return term supplies the next call.

Proof of Theorem B.1. From $U^\dagger U = I$ and $UU^\dagger = I$ we have

$$A^\dagger A + C^\dagger C = I, \qquad B^\dagger B + D^\dagger D = I, \qquad CC^\dagger + DD^\dagger = I.$$

The middle identity gives $\|Dy\|^2 + \|By\|^2 = \|y\|^2$. With $Y_0 = \bigcap_{j \geq 0} \ker BD^j$, it follows that $DY_0 \subseteq Y_0$ and $D|_{Y_0}$ is isometric. In finite dimension it maps Y_0 onto Y_0. Thus U maps $0_X \oplus Y_0$ onto $0_{X'} \oplus Y_0$. Unitarity maps their orthogonal complements $X \oplus Y_1$ and $X' \oplus Y_1$ onto one another. Hence $C(X) \subseteq Y_1$, $D(Y_1) \subseteq Y_1$ for $Y_1 = Y_0^\perp$, and B vanishes on Y_0.

Write $D_1 = D|_{Y_1}$ and $B_1 = B|_{Y_1}$. Put $m = \dim Y$. By Cayley–Hamilton a return vector is invisible exactly when its first m emissions vanish. If $Y_1 = 0$, all delayed exits vanish. Otherwise $m > 0$. For a unit $y \in Y_1$, conservation through m calls gives

$$1 - \|D_1^m y\|^2 = \sum_{j=0}^{m-1} \|BD^j y\|^2 > 0.$$

Equality would make every displayed emission zero and hence put y in Y_0. On the compact unit sphere of Y_1, the continuous function $y \mapsto \|D_1^m y\|$ therefore has a maximum $q < 1$. For $k = pm + r$, $0 \leq r < m$, norm nonincrease on the remaining r calls gives $\|D_1^k\| \leq \|D_1^m\|^p \|D_1^r\| \leq q^p$. This

geometric bound also makes every polynomially weighted exit series converge. Therefore $\sum \|x_k\|$ is finite and $F = (I - D_1)^{-1}C = \sum_j D_1^j C$ solves the wire equation. Every other solution differs by a vector in Y_0 fixed by D, which B kills. The norm identity for $U(u, Fu) = (Lu, Fu)$ proves that L is an isometry. Since $\dim X = \dim X'$ it is onto.

To account for the probability, let $T_j = C^\dagger D^{\dagger j} D^j C$. On a unit input u, $\langle u, T_j u\rangle$ is the probability still inside after call $j+1$. The first block identity gives $x_0^\dagger x_0 = I - T_0$. On later calls the weight that leaves is exactly the drop in the weight still inside, since

$$x_{j+1}^\dagger x_{j+1} = C^\dagger D^{\dagger j} \underbrace{B^\dagger B}_{I - D^\dagger D} D^j C = T_j - T_{j+1}.$$

Adding through exit k gives $\sum_{j=0}^{k} x_j^\dagger x_j = I_X - T_k$. The residual term tends to zero by the decay just proved, giving total exit probability one even for a body with infinitely many possible exits. □

For an internal state w, write $\mathcal{O}w = (BD^j w)_{j\ge 0}$ for the sequence of amplitudes it can still emit. Let $\mathcal{S}(z_0, z_1, \ldots) = (z_1, z_2, \ldots)$ be the left shift. Then $\mathcal{O}D_1 = \mathcal{S}\mathcal{O}$ on Y_1 says that advancing a live state drops its first emission from the sequence. The squared norms of the entries of $\mathcal{O}w$ sum to $\|w\|^2$ for $w \in Y_1$, by the same conservation argument as in the proof. Figure 17 illustrates this with $B = -4/5$ and $D = 3/5$ on the live direction, and with an independent dead direction.

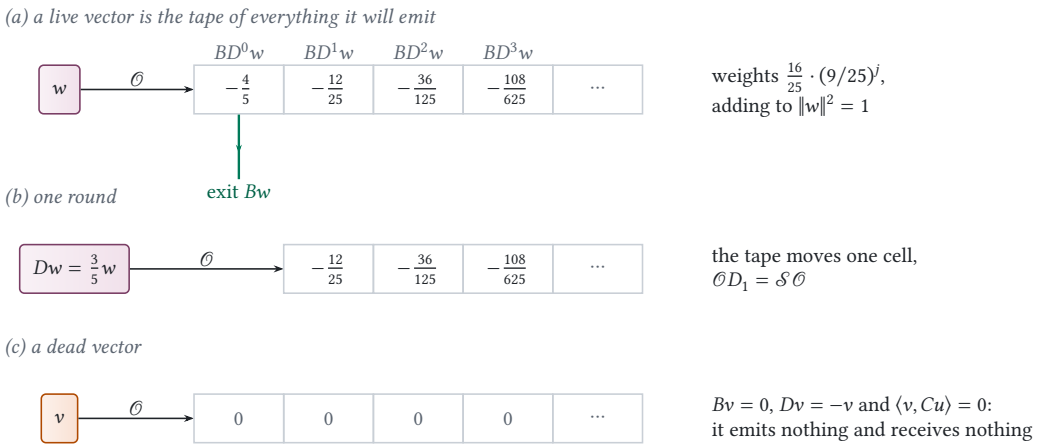


Fig. 17. The future emissions of a live state form its observation tape. Advancing that state shifts the tape by one position. A dead state has the zero tape. Here $Dv = -v$ is dead but not fixed; only the fixed subspace $\ker(I - D)$ obstructs the inverse at phase one.

B.1 Eliminating a pattern

Proposition B.2 (Elimination). *Given the expanded table, eliminating its returning basis values in any order produces a closed iso denoting L in exactly $|\mathcal{B}(c)|$ steps. If the original coefficients lie in a subfield of $\mathbb{C}$, the result has coefficients in that subfield.*

Proof of Proposition B.2. Place the selected return pattern in $Q = \mathbb{C}q$ and put $Y_{\text{rest}} = Y \ominus Q$, the orthogonal complement of Q in Y. The remaining input and output are $E_{\text{in}} = X \oplus Y_{\text{rest}}$ and $E_{\text{out}} = X' \oplus Y_{\text{rest}}$. Grouping those remaining ports together gives body blocks

$$\begin{pmatrix} M & b_q \\ c_q & d_q \end{pmatrix}, \qquad M : E_{\text{in}} \to E_{\text{out}}, \quad b_q : Q \to E_{\text{out}}, \quad c_q : E_{\text{in}} \to Q, \quad d_q \in \mathbb{C}.$$

Here $d_q = D_{qq}$ is the selected self-entry. For $w = (u, y_{\mathrm{rest}})$, the row c_q gives $c_q w = (Cu)_q + \sum_{r \neq q} D_{qr} y_r$ from the main-text coordinate equation, while b_q collects the coefficients of y_q in the other outputs. Closing Q asks for a scalar $\lambda = c_q w + d_q \lambda$ and returns $Mw + b_q \lambda$. When $d_q \neq 1$, solving gives $\lambda = c_q w/(1 - d_q)$ and output $(M + b_q c_q/(1 - d_q))w$. When $d_q = 1$, its self-entry has norm one, so every other entry in both its row and column is zero. Hence $b_q = c_q = 0$, any λ solves the wire equation, and the output is Mw. These cases are exhaustive. In particular $d_q = -1$ or $d_q = i$ uses the first branch, with $b_q = c_q = 0$; no divergent geometric series is invoked. This intermediate table is $\mathrm{Tr}^Q(U)$, with E_{in} and E_{out} as its external spaces. Theorem B.1 proves its unitarity in either case.

Figures 3 and 4 apply this calculation to the scalar rotation and the two return patterns of the flip.

After one elimination, solving the remaining feedback equations gives exactly the output of solving all the original equations. Indeed a simultaneous solution restricts to one of the smaller system, and a solution of the smaller system can be extended by a solution for q at those incident values. The external value is independent of these choices by Theorem B.1. Induct on the number of return patterns. Each step removes one and uses only field operations with nonzero denominators. For a phase on the return wire apply the same argument to the modified unitary. No exchange in the order of a conditionally convergent series of individual walks is needed. □

Closed sums for the running rotation. For `guardRound`, both port spaces are $\mathbb{C}$, so the blocks are scalars: $A = D = t = 3/5$, $C = s = 4/5$ and $B = -s$. The general exit formula becomes $x_0 = t$ and $x_{k+1} = (-s)t^k s = -s^2 t^k$. To compare adding these amplitudes with adding their squared magnitudes, follow the first n calls on unit input. Write r_n for the amplitude still returning, ℓ_n for the sum emitted so far, and p_n for the weight of those exits when the guard is measured. The geometric sums give

$$r_n = s t^{n-1},$$

$$\ell_n = t - s^2 \sum_{j=0}^{n-2} t^j = -1 + (1 + t)t^{n-1},$$

$$p_n = t^2 + s^4 \sum_{j=0}^{n-2} t^{2j} = 1 - s^2 t^{2(n-1)}.$$

The sums are empty when $n = 1$. Table 1 follows the resulting emissions, residuals and sums. The squared length $|\ell_n|^2$ can fall and rise as later emissions interfere. The measured halting weight p_n only increases, because each call adds a nonnegative probability.

A table and an unknown gate. There is also no uniform approximation guarantee for an unknown body given only as a gate. For a scalar rotation with cosine t the trace is $t - (1 - t^2)/(1 - t) = -1$ for $-1 < t < 1$, but at the identity rotation it is 1. Put a bypass beside this scalar and the jump is a relative phase detectable by a Hadamard. A fixed finite circuit built by continuously composing the supplied body has a continuous response; eliminating a given clause table can branch on an exact equality [9, Remark B.1].

C Records, measurement and phase

The counter and measured-loop constructions appear in Figures 6, 7 and 8 in the main text. For a common payload $H = X = X' = Y$, the guard-first transport is $F_g = \mathrm{factor}; \sigma_{H,\mathbf{Qubit}} : H + H \leftrightarrow \mathbf{Qubit} \otimes H$. It gives $\widehat{U}_g = [\![F_g]\!] U [\![F_g]\!]^{-1}$. For a counter, $F_{K,g} = d_K^{-1}; (I_{[K]} \otimes F_g)$ puts counter, guard and data in that order, with inverse transport at the output. These are the equal-payload forms of the general constructions in Section 4.

	one call		coherent feedback		guard measured
call n	emitted x_{n-1}	still returning r_n	sum emitted ℓ_n	squared length $\lvert\ell_n\rvert^2$	halted so far p_n
1	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{3}{5}$	0.3600	0.3600
2	$-\frac{16}{25}$	$\frac{12}{25}$	$-\frac{1}{25}$	0.0016	0.7696
3	$-\frac{48}{125}$	$\frac{36}{125}$	$-\frac{53}{125}$	0.1798	0.9171
4	$-\frac{144}{625}$	$\frac{108}{625}$	$-\frac{409}{625}$	0.4282	0.9701
5	$-\frac{432}{3125}$	$\frac{324}{3125}$	$-\frac{2477}{3125}$	0.6283	0.9893
$n \to \infty$	0	0	-1	1	1

Table 1. guardRound on $|*\rangle$. Coherent feedback adds emissions, while guard measurement adds their squared magnitudes. A record keeps emission x_{n-1} with label $n-1$. Only p_n is a halting probability; decimal entries are rounded to four places.

Use the Hilbert space $H_{\mathrm{rec}} = \ell^2(\mathbb{N})$ for an exact record, in the order counter then data. The squared norm of Ru is $\sum_k \|x_k u\|^2 = \|u\|^2$. For the isometric inclusion $J_K : \mathbb{C}^K \to H_{\mathrm{rec}}$ onto the first K labels,

$$\|(J_K \otimes I)R_K - R\| \;\le\; \sum_{k\ge K} \big\||k \bmod K\rangle - |k\rangle\big\|\,\|x_k\| \;\le\; 2\sum_{k\ge K}\|x_k\|.$$

For $k \ge K$, the two distinct orthogonal count labels have distance $\sqrt{2} \le 2$. The bound therefore tends to zero. This argument only compares the operators produced by finite bodies. The map that folds all square-summable sequences into residue classes would be unbounded, as would the map that sums their entries at phase one.

For an input of the form $|u\rangle\langle v|$, tracing the exact record over its orthonormal basis leaves $\sum_k x_k|u\rangle\langle v|x_k^\dagger$. Such operators span the input matrix algebra, giving (4.2). For a finite counter put $G_j = \sum_{k\equiv j \bmod K} x_k$. Discarding that counter instead gives $\sum_{j<K} G_j \rho G_j^\dagger$. As a result, unequal counts congruent modulo K retain cross terms. For the Fourier forms, the coefficient sum is absolutely convergent, and

$$G_j = \frac{1}{K}\sum_{z^K=1} \bar{z}^{\,j}\Theta(z), \qquad x_k = \frac{1}{2\pi}\int_0^{2\pi} e^{-ik\varphi}\Theta(e^{i\varphi})\,d\varphi.$$

Expanding $\Theta(z)\rho\Theta(z)^\dagger$ and integrating term by term removes all unequal indices, proving (6.1).

The table at the end of Section 4 gives these maps for bypassRound. For the classical trace, let $\theta(\rho,\sigma) = \mathrm{diag}(\rho,\sigma)$ insert the two operator blocks and let φ extract the diagonal output blocks. Then $\Phi_U = \varphi \circ \mathcal{E}_U \circ \theta$ is the branching map of Section 4. Its trace on completely positive trace-nonincreasing maps on finite C^*-algebras is the unrolling $\sum_k x_k \rho x_k^\dagger$ [3, Prop. 3.2.17].

Relation to monitored first returns. The published test-first loop measures before applying U [26, § 3]. To obtain our body-first convention when $X' = X$, apply U once before entering it. Let $\iota_X u = (u, 0)$ and $P = \iota_X \iota_X^\dagger$. For $n \ge 1$, the exit map on call n is $\iota_X^\dagger U((I-P)U)^{n-1}\iota_X = x_{n-1} : X \to X$. Thus the monitored-return count includes the initial call and is one more than our number of returns [6, § 2].

D Agreement and replacement

Proof of Theorem 5.1. Assume $\mathcal{M} = \mathcal{E}_L$. For each unit input u, test the output on any w orthogonal to Lu. The sum of the nonnegative numbers $|\langle w, x_k u\rangle|^2$ is zero. Thus $x_k u$ is parallel to

Lu for each k and all u. Put $V = L^\dagger x_k$. Since Vu lies along u, testing a basis vector e_i gives $Ve_i = \lambda_i e_i$. To see why the scalar cannot depend on i, test two distinct basis vectors together:

$$V(e_i + e_j) = \underbrace{\lambda_i e_i + \lambda_j e_j}_{\text{by linearity}} = \underbrace{\lambda_{ij}(e_i + e_j)}_{\text{parallel to the input}} .$$

Comparing the two coordinates gives $\lambda_i = \lambda_{ij} = \lambda_j$. Hence $V = b_k I$ and $x_k = b_k L$, including in dimension one where there is only one basis coefficient to begin with. By absolute convergence, $\sum_k b_k = 1$ since $\sum_k x_k = L \neq 0$. By $\sum_k x_k^\dagger x_k = I$, $\sum_k |b_k|^2 = 1$. The unit vector $\xi = \sum_k b_k |k\rangle$ therefore satisfies $Ru = \xi \otimes Lu$. Conversely that factorization implies $\mathcal{M} = \mathcal{E}_L$ after tracing out the count; testing each label shows $x_k = \langle k|\xi\rangle L$.

These conditions imply that $x_0, \dots, x_m$ span at most a line. If $m > 0$, Cayley–Hamilton writes D^m as a linear combination of $I, D, \dots, D^{m-1}$. Since $x_{k+1} = BD^k C$, multiplication by higher powers shows that every exit after x_m belongs to the span of $x_1, \dots, x_m$. If $m = 0$, all exits after x_0 vanish. Thus the span of the first $m+1$ exits is the span of all the exits. It cannot be zero because it contains their unitary sum on the nonzero input. If it is a line, the same sum L spans it, so $x_k = b_k L$ for each k. Completeness of the exits again gives the unit norm of ξ. This proves all implications. □

Channel contexts. Fix the hole type $a \rightsquigarrow b$ and abbreviate it by $\mathcal{H}$ to the left of a context judgment. For fixed closed channels h, h_j, the contexts are generated by

$$C ::= [-] \mid h; C \mid C; h \mid h \otimes C \mid C \otimes h \\ \mid \mathrm{case}\big(\Psi_i; C, (\Psi_j; h_j)_{j\neq i}\big).$$

The rules for the hole, sequence and tensor are

$$\mathcal{H} \vdash [-] : a \rightsquigarrow b \qquad \frac{h : e \rightsquigarrow c \qquad \mathcal{H} \vdash C : c \rightsquigarrow d}{\mathcal{H} \vdash h; C : e \rightsquigarrow d} \qquad \frac{\mathcal{H} \vdash C : c \rightsquigarrow d \qquad h : d \rightsquigarrow e}{\mathcal{H} \vdash C; h : c \rightsquigarrow e}$$

$$\frac{h : e \rightsquigarrow f \qquad \mathcal{H} \vdash C : c \rightsquigarrow d}{\mathcal{H} \vdash h \otimes C : e \otimes c \rightsquigarrow f \otimes d} \qquad \frac{\mathcal{H} \vdash C : c \rightsquigarrow d \qquad h : e \rightsquigarrow f}{\mathcal{H} \vdash C \otimes h : c \otimes e \rightsquigarrow d \otimes f}.$$

For a finite instrument $(\Psi_j : c \to c_j)_{j\in J}$, meaning $\Psi_j : \mathrm{End}(\llbracket c \rrbracket) \to \mathrm{End}(\llbracket c_j \rrbracket)$, the maps are completely positive and their output traces sum to the input trace. Its case rule is

$$\frac{i \in J \quad \mathcal{H} \vdash C : c_i \rightsquigarrow d \quad h_j : c_j \rightsquigarrow d \ (j \neq i)}{\mathcal{H} \vdash \mathrm{case}\big(\Psi_i; C, (\Psi_j; h_j)_{j\neq i}\big) : c \rightsquigarrow d}.$$

Plugging h into the hole interprets sequencing by channel composition and tensor by the tensor product of channels. The case denotes $\llbracket C[h] \rrbracket \circ \Psi_i + \sum_{j\neq i} \llbracket h_j \rrbracket \circ \Psi_j$. Thus exactly one syntactic occurrence depends on the hole, and each branch continuation has the channel type stated by the rules. These contexts can use a hole on one instrument outcome and omit it on the others. They do not contain iso-level operations on that hole.

Proof of Corollary 5.2. If $X = 0$ then $X' = 0$ as well, and there is just one map on the zero operator space. Now suppose $X \neq 0$. On agreement the finite record has the form $R_K u = \xi_K \otimes Lu$, where $\xi_K = \sum_k b_k |k \bmod K\rangle$. Since R_K is an isometry and an input unit vector exists, $\|\xi_K\| = 1$, even if components with equal labels interfere. Discarding it gives $\mathcal{E}_L$. For the exact record the equality after discard needs no agreement condition. Channel equality on data entails equality after tensoring with identity on any finite reference, since elementary tensor operators span the joint matrix algebra. Composition with fixed channels, discarding and applying a fixed instrument on either program therefore preserve equality. For a measurement-conditioned continuation apply the

same argument separately to each unnormalized outcome and then add if the outcome is forgotten. This proves the claim by induction over a context built with these constructors. Conversely, if the channels differ, some density operator distinguishes them since positive trace-one inputs span the matrix algebra over $\mathbb{C}$. The two resulting density operators are distinguished by a spectral projection of their Hermitian difference. This test is at the output of the empty context; finite exact isos and tag measurement can implement a measurement in the chosen eigenbasis.

For delayed discard, use the following separate semantic continuation calculus. Its ordinary data types are finite, while its opaque history names may carry countable semantic record spaces. Write $\Gamma \vdash K : a \Rightarrow b$ for a finite continuation that starts with ordinary data of type a and the fresh opaque history names in Γ, and returns data of type b after consuming all those names. Each name carries its designated record space. Each branch Ψ_j of a finite instrument sends operators on $[\![a]\!]$ to operators on $[\![a_j]\!]$ by a completely positive map. The traces of these outputs sum to one on each density input. An ordinary channel is its one-outcome case. For a binder, let $R : [\![a]\!] \to H \otimes [\![d]\!]$ be a recording isometry. Its history name g has space H. The caller rules are

$$\varnothing \vdash \mathsf{return} : a \Rightarrow a \qquad \frac{\Gamma \vdash K_j : a_j \Rightarrow b \quad (j \in J)}{\Gamma \vdash \mathsf{case}\ (\Psi_j; K_j)_{j\in J} : a \Rightarrow b}$$

$$\frac{\Gamma \vdash K : a \Rightarrow b \quad h \notin \Gamma}{\Gamma, h \vdash \mathsf{drop}\ h; K : a \Rightarrow b} \qquad \frac{\Gamma, g \vdash K : d \Rightarrow b \quad g \notin \Gamma}{\Gamma \vdash \mathsf{record}\ g = R\ \mathsf{in}\ K : a \Rightarrow b}.$$

Bound names are chosen fresh throughout the finite derivation by renaming. For an ordering of the live names, let H_Γ be their tensor product and I_Γ its identity. The semantics maps trace-class operators on $H_\Gamma \otimes [\![a]\!]$ to operators on $[\![b]\!]$. With canonical reassociations and permutations of named factors understood, its equations are

$$\begin{aligned}
[\![\mathsf{return}]\!] &= \mathrm{id},\\
[\![\mathsf{case}\ (\Psi_j; K_j)_{j\in J}]\!] &= \sum_{j\in J} [\![K_j]\!] \circ (\mathcal{E}_{I_\Gamma} \otimes \Psi_j),\\
[\![\mathsf{drop}\ h; K]\!] &= [\![K]\!] \circ \mathrm{tr}_h,\\
[\![\mathsf{record}\ g = R\ \mathsf{in}\ K]\!] &= [\![K]\!] \circ \mathcal{E}_{I_\Gamma \otimes R}.
\end{aligned}$$

The instrument selects a continuation by its classical outcome and passes all live names to that continuation. A binder adds its new record, and a drop traces out just the named factor. Only the empty history context can return, so each returning branch drops every live name exactly once. For K_h starting with a distinguished h, fresh for every internal record binder, delete its drops to obtain K, leaving all other binders and drops in place. The rules show that K is well typed with h removed. On each returning branch and for any joint trace-class input τ,

$$[\![K_h]\!](\tau) = [\![K]\!](\mathrm{tr}_h\, \tau).$$

For a drop it is the definition. For an ordinary operation Ψ before the drop, use $\mathrm{tr}_h((\mathcal{E}_{I_h} \otimes \Psi)(\tau)) = \Psi(\mathrm{tr}_h\, \tau)$, by taking the partial trace in a basis of h. This holds for each subnormalized instrument outcome, and tensoring with a reference leaves it valid. Nested recording is an isometry on the ordinary data and the new record, with the older name carried by the identity. Partial traces over distinct names commute at their eliminations. Induction over sequencing, measurement branches and nested bindings proves the formula. Consequently a recording atom R followed by K_h may be replaced by its marginal channel $\rho \mapsto \mathrm{tr}_H(R\rho R^\dagger)$ followed by K. When R is the exact exit record of our body β, this marginal is $\mathsf{while}_c\, \beta$ by (4.2). Under agreement it may also be replaced by the embedded coherent iso. A finite clock has its own marginal and does not in general admit the

unconditional exact-record replacement. If a caller is a limit of increasing finite executions with subnormalized outputs, the same pathwise identities preserve its lost trace under that limit. This last observation assumes such a semantics; general recursive callers are outside the finite grammar. □

Why one finite clock does not suffice. The output interface matters. If a caller reads the count, even a fixed count independent of the data is an observable output. Conversely, a finite clock may hide a failure of agreement. Use the permutation $E0 \mapsto E0$ and $E1 \mapsto Bq_1 \mapsto \cdots \mapsto Bq_K \mapsto E1$, with K distinct return values $q_1, \dots, q_K$. Its only nonzero exits are $x_0 = |0\rangle\langle 0|$ and $x_K = |1\rangle\langle 1|$. The K-clock assigns label 0 to both exits. Ignoring it preserves $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, whereas a measurement on every return dephases $|+\rangle$ in that basis. Checking just this clock would accept the replacement incorrectly.

E The derivative and its average

Proof of Theorem 6.1. Work on Y_1. The geometric bound in Theorem B.1 gives an open disk of radius greater than one on which F_z and $\Theta(z)$ are analytic. Empty live space gives $F_z = 0$. The identity $F_z = C + zD_1F_z$ says

$$U_1(u, zF_zu) = (\Theta(z)u, F_zu).$$

Use the wire equation twice, first at (v, w) and then at (u, z). Because $U_1 : X \oplus Y_1 \to X' \oplus Y_1$ preserves inner products, the inner product of the inputs equals that of the outputs. With the inner product linear in its second argument, the two sides are

$$\underbrace{\langle v, u\rangle + \bar{w}z\langle F_wv, F_zu\rangle}_{\text{external and returning inputs}} = \underbrace{\langle \Theta(w)v, \Theta(z)u\rangle + \langle F_wv, F_zu\rangle}_{\text{external and returning outputs}}.$$

Collecting terms gives $I - \Theta(w)^\dagger\Theta(z) = (1 - \bar{w}z)F_w^\dagger F_z$. Keep w fixed on the unit circle and differentiate with respect to z. At $z = w$ the factor $1 - \bar{w}w$ is zero, so the term containing F_z' vanishes. The remaining equality is $\Theta(w)^\dagger\Theta'(w) = \bar{w}F_w^\dagger F_w$. Now put $w = e^{i\varphi}$. The angular derivative multiplies $\Theta'(w)$ by iw, giving $Q(w) = F_w^\dagger F_w$ as claimed.

By the telescope in Appendix B, $\sum_{k=1}^{p} kx_k^\dagger x_k = \sum_{j=0}^{p-1} T_j - pT_p$. Geometric decay makes $pT_p \to 0$. Thus $N = \sum_j T_j$. In $F_z^\dagger F_z$ the terms at m, n have factor z^{n-m} on the unit circle. As in Figure 10, integration keeps only the cells where $m = n$. Uniform absolute convergence permits this termwise integration, leaving $\sum_j T_j$. This gives the phase average. For the trace of N, cyclicity changes the summand $\operatorname{tr} T_j$ into $\operatorname{tr}(D_1^j CC^\dagger D_1^{\dagger j})$. From the Y_1 output block of $U_1U_1^\dagger = I_{X'\oplus Y_1}$ we have $CC^\dagger = I - D_1D_1^\dagger$. The partial sum once again telescopes,

$$\sum_{j=0}^{p-1} D_1^j CC^\dagger D_1^{\dagger j} = I_{Y_1} - D_1^p D_1^{\dagger p} \longrightarrow I_{Y_1}.$$

Thus $\operatorname{tr} N = \dim Y_1$. On each unit input u the measured probability of exit k is $\|x_ku\|^2$, so its mean count is $\sum_k k\|x_ku\|^2 = \langle u, Nu\rangle$. □

For the monitored first-return convention the generating function is $z\Theta(z)$, whose relative derivative is $I + Q(z)$. Its mean number of calls is therefore $1 + \langle u, Nu\rangle$ [6, § 3, Eq. (8)]. A phase on the return tag is an ordinary iso, so forming this family requires no counter or guard measurement.

Phase invariance and agreement. On a nonempty external type, the phase family also characterizes agreement. If every exit is $x_k = b_kL$, then $\Theta(z) = f(z)L$ with $|f(z)| = 1$ on the circle, so every marking phase gives the same data channel. Conversely, if those channels are all equal, they equal

$\mathcal{E}_L$ because $\Theta(1) = L$. Their average is $\mathcal{M}$ by (6.1), so $\mathcal{M} = \mathcal{E}_L$. Thus agreement says exactly when changing the return phase leaves the data channel unchanged.

F Clause programs and their observations

Many small loop bodies can be written as two clauses with a variable for the payload. We give their definitions here so that one can reproduce each diagram directly from an iso. Each program closes its returning summand with **trace**. We abbreviate **exit** and **back** by E and B in this section's source equations. In these equations B is an injection tag; a matrix block appears only in a denotational equation. In executable types, `x` spells tensor product, and a coefficient `(a :+ b)` denotes $a + ib$ for real a, b.

Two scalar loops in sequence. Use two copies of the scalar guard rotation, with $t = 3/5$ and $s = 4/5$. Their separate returns each receive the phase z, and their external ports connect in sequence. The responses therefore multiply to $f(z)^2$, where $f(z) = (t - z)/(1 - tz)$. Figure 18 gives both this cascade and its single-round matrix W on $\mathbb{C} \oplus (\mathbb{C} \oplus \mathbb{C})$. Multiplying its blocks, or solving its two return equations, gives the same $f(z)^2$. The columns of W give the following three clauses; its two returning basis values are `back 0` and `back 1`.

```
round twoSections : Unit + Qubit = {
  exit unit ↔ [9/25 . exit unit + 4/5 . back 0 + 12/25 . back 1];
  back 0 ↔ [-12/25 . exit unit + 3/5 . back 0 + -16/25 . back 1];
  back 1 ↔ [-4/5 . exit unit + 3/5 . back 1]
}

trace twoSections on unit
```

The command returns $|*\rangle$, since $f(1)^2 = 1$. A period-two clock also returns label 0: both $f(1)^2$ and $f(-1)^2$ are one, so the odd residue sum is zero.

Now carry data u unchanged through both sections and apply a gate L once on the final external output. As Figure 19 shows, this amounts to tensoring the internal evolution with the identity on the data. The response is $f(z)^2 L$. Every exit coefficient multiplies the same L, so dropping the count preserves the data channel by Theorem 5.1. On the unit circle $|f| = 1$, hence the relative derivative of the squared scalar response is twice that of one section. Its phase average gives mean count two, as expected from the mean of one for each rotation. Changing coordinates on the internal space or adjoining an inaccessible summand leaves the exits unchanged.

F.1 A flip on every call

For a loop that also transforms a payload on every call, take a guard and a data qubit, apply R_θ to the guard and the data flip $J = \mathsf{X}$ to the data, and close the returning summand. This is more structure than the scalar loop of the main text needs, but it gives a second example where measured and coherent outputs differ. The following source binds the data in each clause before applying its gate.

```
X :: Qubit ↔ Qubit
X = { 0 ↔ 1; 1 ↔ 0 }

round flipRound : Qubit + Qubit = {
  exit x ↔ let y = X x in [ 3/5 . exit y + 4/5 . back y];
  back x ↔ let y = X x in [-4/5 . exit y + 3/5 . back y]
}
```

Two copies of the scalar rotation, $t = 3/5$, $s = 4/5$



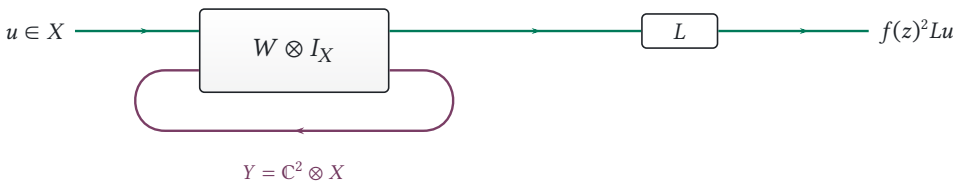
$$f(z) = \frac{t - z}{1 - tz}, \qquad f(1)^2 = 1$$

$$W = \begin{pmatrix} 9/25 & -12/25 & -4/5 \\ 4/5 & 3/5 & 0 \\ 12/25 & -16/25 & 3/5 \end{pmatrix}$$

The same cascade as one body on $\mathbb{C} \oplus (Y_1 \oplus Y_2)$.
Its external-first blocks give $\Theta_W(z) = f(z)^2$.

Fig. 18. Two specified scalar sections give response $f(z)^2$. The numerical matrix is the open body of this cascade in external-first order.

Carry data through this two-section example

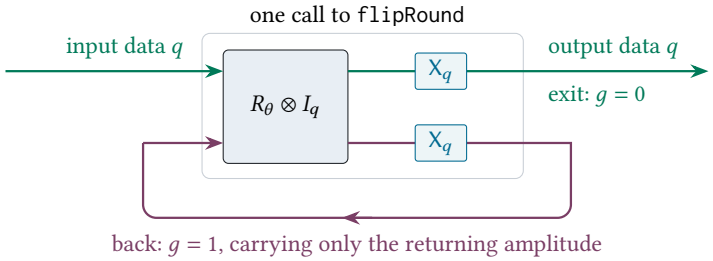


$$U = (L \oplus I_Y)(W \otimes I_X), \qquad \Theta_U(z) = f(z)^2 L$$

Here W is the displayed rational two-section body and $L : X \to X$ is unitary. Each exit is a scalar times L. Dropping its count preserves the data channel. The mean number of returns is two on every unit input.

Fig. 19. Tensor the specified two-section body with the data identity and apply L on its external output. This constructed body passes agreement and has mean count two.

The first definition names the data gate X inside the box in Figure 20. On input `exit 0` the first clause binds $x = 0$, so `let y = X x` binds $y = 1$. Each dot scales a tagged output, and the brackets form their superposition. The coefficients $3/5$ and $4/5$ are the first column of R_θ, since **`exit`** means that the guard enters at 0. Following `back 1` into the second clause flips the data once more and uses the other column, with coefficients $-4/5$ and $3/5$. This is where the second call acquired its minus sign. With the guard first and the data second, Figure 21 is the matrix of $R_\theta \otimes \mathrm{X}$.



Call	Component entering this call	Leaves the loop	Continues to the next call
1	$\|00\rangle$	$\frac{3}{5}\|01\rangle$	$\frac{4}{5}\|11\rangle$
2	$\frac{4}{5}\|11\rangle$	$-\frac{16}{25}\|00\rangle$	$\frac{12}{25}\|10\rangle$
3	$\frac{12}{25}\|10\rangle$	$-\frac{48}{125}\|01\rangle$	$\frac{36}{125}\|11\rangle$

Fig. 20. The coherent loop connects the body's returning output to its returning input. The two ports are alternative guard values, each carrying the same data qubit q. Both alternatives apply X_q; this data gate commutes with the guard rotation. The table starts at $|00\rangle$ and writes joint states $|gq\rangle$, with guard first. Emitted components do not enter a later row; only the last column does. No guard measurement is made here.

```
exit 0 <-> [3/5 . exit 1 + 4/5 . back 1]
exit 1 <-> [3/5 . exit 0 + 4/5 . back 0]
back 0 <-> [-4/5 . exit 1 + 3/5 . back 1]
back 1 <-> [-4/5 . exit 0 + 3/5 . back 0]
```

one column for each ground input

input basis patterns

outputs	exit 0	exit 1	back 0	back 1
exit 0	A 0	$\frac{3}{5}$	0	$-\frac{4}{5}$ B
exit 1	$\frac{3}{5}$	0	$-\frac{4}{5}$	0
back 0	0	$\frac{4}{5}$	0	$\frac{3}{5}$
back 1	C $\frac{4}{5}$	0	$\frac{3}{5}$	0 D

Compare inputs `exit 0` and `back 0`

output row	column product
`exit 1`	$(3/5)(-4/5)$
`back 1`	$(4/5)(3/5)$
inner product	0

The other rows contribute zero.
Each squared norm is $9/25 + 16/25 = 1$.

Fig. 21. The four columns of the payload-flipping body, in external-first order. This is $R_\theta \otimes \mathrm{X}$.

The exit maps of `flipRound`. Put $t = 3/5$ and $s = 4/5$. The two clauses of `flipRound` give $A = t\mathrm{X}$, $C = s\mathrm{X}$ on entry and $B = -s\mathrm{X}$, $D = t\mathrm{X}$ on return. Hence $x_0 = t\mathrm{X}$ and $x_1 = BC = -s^2 I$. The data flips twice before that second exit. In general $x_{k+1} = -s^2 t^k \mathrm{X}^{k+2}$. Grouping equal data gates gives two

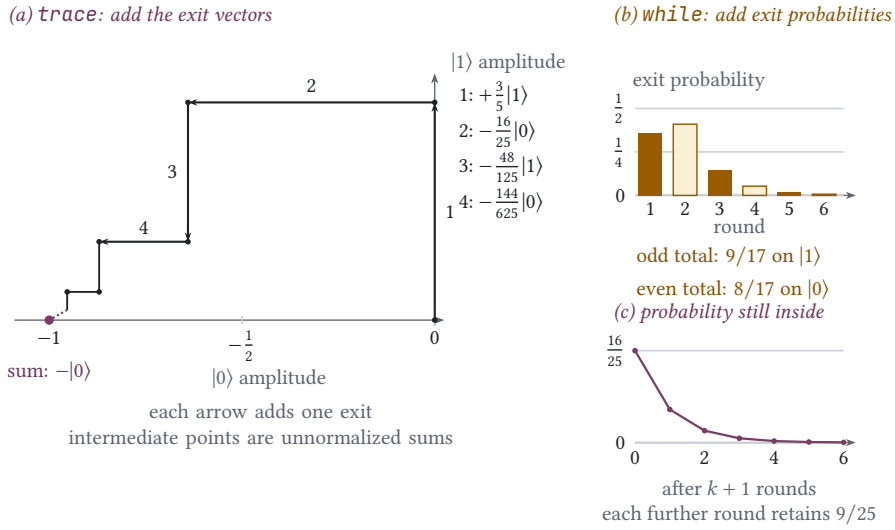


Fig. 22. On input $|0\rangle$, the left panel adds exit amplitudes in the data space. The right panel keeps the exit rounds distinct and plots their probabilities. Calls are numbered from one.

ordinary geometric series,

$$\underbrace{t - s^2 t \sum_{j\geq 0} t^{2j} = 0,}_{\text{coefficient of X}} \qquad \underbrace{-s^2 \sum_{j\geq 0} t^{2j} = -1,}_{\text{coefficient of } I} \qquad s^2 = 1 - t^2.$$

Thus the whole coherent loop returns $-u$. In particular it can never return data 1 on input 0. With an exact count the exits that return 0 on this input have odd counts, with amplitudes $-s^2, -s^2t^2, \ldots$. Their squared magnitudes sum to $s^4/(1-t^4) = 8/17$. The complementary probability $9/17$ goes to 1. These are the two observations in Figure 22.

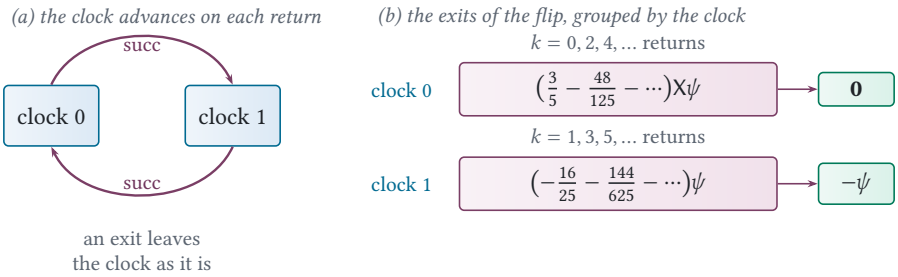


Fig. 23. For the flip, folding by parity cancels all even counts as amplitudes. Measurement of an exact count has positive probability at every count.

The numerical elimination also gives a finite calculation of this loop. Unlike `guardRound`, it has two returning basis patterns. Figure 4 in the main text shows the remaining clauses after each elimination.

The first substitution in Figure 4 removes `back 0` and leaves coefficient $(3/5)^2 = 9/25$ on the remaining return amplitude y_1. It also changes the coefficient of y_1 in external output v_1 to $(3/5)(-4/5) = -12/25$. On external basis input 0, an amplitude $4/5$ enters this remaining return pattern. Solving its scalar wire equation gives the two entries in that output column:

$$\text{on data } 0: \quad \frac{(4/5)(-4/5)}{1 - 9/25} = -1,$$

$$\text{on data } 1: \quad \frac{3}{5} + \frac{(4/5)(-12/25)}{1 - 9/25} = 0.$$

Interchanging the data bits gives the other column. Thus the finite rewrite produces the same $-I$ that we obtained by adding the exits round by round.

Moving the flip to the exit. A variant that passes the agreement test has the circuit in Figure 24. In the ground clauses, the bit after **exit** is the flipped input bit and the bit after **back** is the input itself.

```
round flipOnExit : Qubit + Qubit = {
  exit 0 ↔ [ 3/5 . exit 1 + 4/5 . back 0];
  exit 1 ↔ [ 3/5 . exit 0 + 4/5 . back 1];
  back 0 ↔ [-4/5 . exit 1 + 3/5 . back 0];
  back 1 ↔ [-4/5 . exit 0 + 3/5 . back 1]
}
```

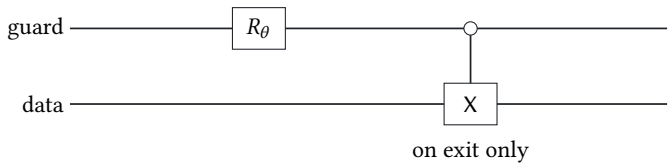


Fig. 24. The same rotation as in `flipRound`, with the flip controlled on the outgoing guard being 0. A returning component keeps its data. Whichever round exits, that component of the data is flipped exactly once.

Running `trace flipOnExit on 0` gives $-|1\rangle$. The measured command `while flipOnExit on 0` gives $|1\rangle\langle 1|$. Its exits are $(3/5)\mathsf{X}$ at count zero and $-(16/25)(3/5)^{k-1}\mathsf{X}$ at counts $k \geq 1$, using the scalar coefficients calculated in Section 3. Every exit is proportional to X, so agreement holds on every input, including one entangled with a reference. The coherent completion is $-\mathsf{X}$.

Rotation with a data iso. For $c = \cos\theta$, $s = \sin\theta$ and an iso $M : a \leftrightarrow a$, define $\mathrm{rotbody}(\theta, M) : a + a \leftrightarrow a + a$ by the clauses

$$E\,x \leftrightarrow \text{let } y = M\,x \text{ in } c\,E\,y + s\,B\,y,$$

$$B\,x \leftrightarrow \text{let } y = M\,x \text{ in } -s\,E\,y + c\,B\,y.$$

The notation is a schema for named first-order isos, not a higher-order value. Putting $M = \mathsf{X}$ gives the four ground instances of the flip. For $|c| < 1$ its exits are cM and $-s^2c^kM^{k+2}$. Summing them gives

$$f_\theta(M) = M(cI - M)(I - cM)^{-1}.$$

On an eigenvector of M with eigenvalue w, the equation $f_\theta(w) = 1$ reduces to $w^2 - 2cw + 1 = 0$, while $f_\theta(w) = -1$ reduces to $w^2 = 1$. The eigenvalues of $R_{-\theta}$ are $c \mp is$, the roots of the first polynomial, so $f_\theta(R_{-\theta}) = I$. When $|c| = 1$, the guard never enters and the immediate exit gives the same conclusion for that rotation. The eigenvalues of X are ± 1, so $f_\theta(\mathsf{X}) = -I$ for $|c| < 1$.

Here is the inverse-rotation instance at $c = 3/5$, including its invocation. The first declaration supplies $M = R_{-\theta}$; the second applies it locally on each call.

```
inverseRotation :: Qubit ↔ Qubit
inverseRotation = {
  0 ↔ [3/5 . 0 + -4/5 . 1];
  1 ↔ [4/5 . 0 + 3/5 . 1]
}

round undoRound : Qubit + Qubit = {
  exit x ↔ let y = inverseRotation x in
    [3/5 . exit y + 4/5 . back y];
  back x ↔ let y = inverseRotation x in
    [-4/5 . exit y + 3/5 . back y]
}

trace undoRound on 0
```

The command returns $|0\rangle$. On an arbitrary vector the completed gate is I, as the eigenvalue calculation shows. The measured loop need not agree: on $|+\rangle$ it gives $\frac{1}{650}\begin{pmatrix} 349 & -107 \\ -107 & 301 \end{pmatrix}$ rather than $|+\rangle\langle+|$. The cancellation giving the coherent identity takes place between exits on different rounds.

A guard whose two payloads differ. A tag test need not be a measurement of a qubit register beside an unchanged data type. Take the two external patterns 0, 1 and just one return pattern $*$. The body below is the three-cycle of monitored recurrence, with its basis relabeled [6, Example 3.1]. It binds no history variable.

```
cycle3 :: (Qubit + Unit) ↔ (Qubit + Unit)
cycle3 = {
  left (left unit)  ↔ right unit;
  left (right unit) ↔ left (left unit);
  right unit        ↔ left (right unit)
}
```

Input 1 exits immediately as 0, whereas input 0 returns once and exits as 1. Thus $x_0 = |0\rangle\langle 1|$, $x_1 = |1\rangle\langle 0|$, and all later exits vanish. Calling **trace** on this iso gives the flip X, so $|+\rangle$ returns unchanged. Calling **count** with period two gives $(|0\rangle|0\rangle + |1\rangle|1\rangle)/\sqrt{2}$, with counter first. Calling **while** on the very same iso returns $I/2$ on that input. The two exit vectors on $|+\rangle$ have zero inner product, but they are values of the data we retain. Coherent feedback still preserves the terms $|0\rangle\langle 1|$ and $|1\rangle\langle 0|$ between them. A separate ignored count removes these terms because it gives the two data vectors different orthogonal labels. Thus distinguishable data values alone do not make recording the round harmless.

A cycle, a reflection and a phase. The cycle3 program just defined gives the data flip after closing its one return pattern. One may add a dead pattern with its own identity clause, obtaining the same flip on a two-pattern return type. A body with no data can instead be the reflection $E * \leftrightarrow t\,E\ * + s\,B *$, $B * \leftrightarrow s\,E\ * - t\,B *$ at $t = 3/5, s = 4/5$. Its immediate exit is t, and its delayed sum is $s^2/(1+t)$. They add to 1.

```
round reflection : Unit + Unit = {
  exit unit ↔ [3/5 . exit unit + 4/5 . back unit];
  back unit ↔ [4/5 . exit unit + -3/5 . back unit]
}

trace reflection on unit
```

This command returns $|*\rangle$. The measured loop also returns the sole external value with probability one. Its parity record is $|1\rangle \otimes |*\rangle$: the even exits cancel and the odd exits add to one. This reflection and `guardRound` therefore have the same measured data output but opposite coherent phases.

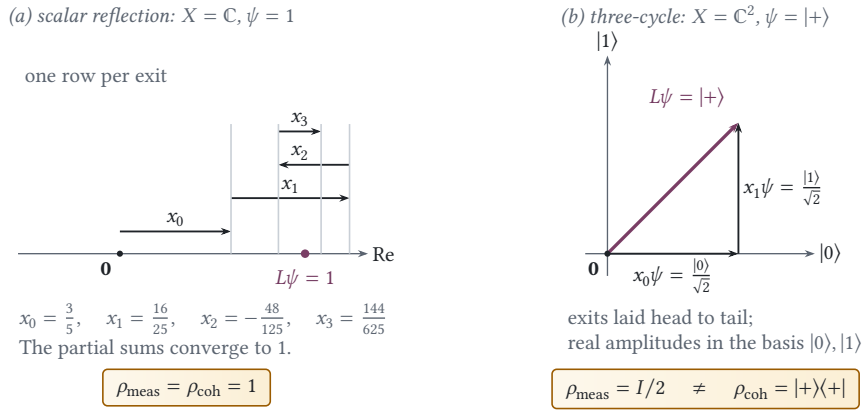


Fig. 25. The scalar reflection gives the same data under measurement or coherent feedback. For the three-cycle, measuring yields $I/2$ while coherent feedback returns the pure state $|+\rangle$.

For another example, take the four permutation clauses

$$E0 \mapsto E0, \quad E1 \mapsto B1, \quad B0 \mapsto B0, \quad B1 \mapsto E1.$$

Multiply every output by g with $|g| = 1$. Input 0 receives one factor of g and input 1 receives two, giving $\mathrm{diag}(g, g^2)$. The one-call channel is unchanged by this common phase but the loop channel in general changes. Placing a phase z on the *returning* outputs alone instead gives $\mathrm{diag}(1, z)$. The markings occur in different places, but at $z = g$ their completions differ only by the global factor g. They therefore give the same data channel for this permutation. That channel can still differ from the unmarked identity channel.

The following two programs instantiate $g = z = i$, written `(0 :+ 1)` in executable syntax. Both use the exact input $3|0\rangle/5 + 4|1\rangle/5$.

```
round unmarked : Qubit + Qubit = {
  exit 0 ↔ exit 0;
  exit 1 ↔ back 1;
  back 0 ↔ back 0;
  back 1 ↔ exit 1
}

round wholePhase : Qubit + Qubit = {
  exit 0 ↔ (0 :+ 1) . exit 0;
  exit 1 ↔ (0 :+ 1) . back 1;
  back 0 ↔ (0 :+ 1) . back 0;
  back 1 ↔ (0 :+ 1) . exit 1
}

trace wholePhase on 3/5 . 0 + 4/5 . 1
```

This invocation returns $3i|0\rangle/5 - 4|1\rangle/5$.

```
round returnPhase : Qubit + Qubit = {
  exit 0 ↔ exit 0;
  exit 1 ↔ (0 :+ 1) . back 1;
  back 0 ↔ (0 :+ 1) . back 0;
  back 1 ↔ exit 1
}

trace returnPhase on 3/5 . 0 + 4/5 . 1
```

The second returns $3|0\rangle/5 + 4i|1\rangle/5$. Multiplying it by i gives the first output, so their density operators agree.

Kickback with a returned ancilla. Let $J = \mathsf{X}$, H be Hadamard, $G = HJ$, and $\Pi_\pm = (I \pm J)/2$. Use $\mathrm{rotbody}(\pi/4, M)$ with ancilla and data as payload, where $M = G \otimes \Pi_+ + JG \otimes \Pi_-$. In iso syntax this M is an operation controlled in the J basis of the data: change that basis with H, apply G or JG to the ancilla according to the resulting tag, then undo the basis change. In the first sector $G^2 = \sqrt{2}G - I$, so $f_{\pi/4}(G) = I$. In the second $(JG)^2 = I$, so $f_{\pi/4}(JG) = -I$. The completed loop is $I \otimes (\Pi_+ - \Pi_-) = I \otimes J$. Thus it returns the ancilla on arbitrary joint inputs, not only on a selected initialization.

In the basis of the two payload bits, the projector sum gives

$$M|0, d\rangle = \frac{|0, 1-d\rangle - |1, 1-d\rangle}{\sqrt{2}}, \qquad M|1, d\rangle = \frac{|0, d\rangle + |1, d\rangle}{\sqrt{2}}.$$

The data bit flips in the first case and stays in the second. Expanding the guard rotation now gives the clauses below. Each nonzero coefficient of M is $\pm 1/\sqrt{2}$, and the guard contributes another $1/\sqrt{2}$, leaving only coefficients $\pm 1/2$. Thus no rounded Hadamard coefficient is needed in this source.

```
round kickbackExact : (Qubit x Qubit) + (Qubit x Qubit) = {
  exit ⟨0, 0⟩ ↔ [1/2 . exit ⟨0, 1⟩ + -1/2 . exit ⟨1, 1⟩
                + 1/2 . back ⟨0, 1⟩ + -1/2 . back ⟨1, 1⟩];
  exit ⟨0, 1⟩ ↔ [1/2 . exit ⟨0, 0⟩ + -1/2 . exit ⟨1, 0⟩
                + 1/2 . back ⟨0, 0⟩ + -1/2 . back ⟨1, 0⟩];
  exit ⟨1, 0⟩ ↔ [1/2 . exit ⟨0, 0⟩ + 1/2 . exit ⟨1, 0⟩
                + 1/2 . back ⟨0, 0⟩ + 1/2 . back ⟨1, 0⟩];
  exit ⟨1, 1⟩ ↔ [1/2 . exit ⟨0, 1⟩ + 1/2 . exit ⟨1, 1⟩
                + 1/2 . back ⟨0, 1⟩ + 1/2 . back ⟨1, 1⟩];
  back ⟨0, 0⟩ ↔ [-1/2 . exit ⟨0, 1⟩ + 1/2 . exit ⟨1, 1⟩
                + 1/2 . back ⟨0, 1⟩ + -1/2 . back ⟨1, 1⟩];
  back ⟨0, 1⟩ ↔ [-1/2 . exit ⟨0, 0⟩ + 1/2 . exit ⟨1, 0⟩
                + 1/2 . back ⟨0, 0⟩ + -1/2 . back ⟨1, 0⟩];
  back ⟨1, 0⟩ ↔ [-1/2 . exit ⟨0, 0⟩ + -1/2 . exit ⟨1, 0⟩
                + 1/2 . back ⟨0, 0⟩ + 1/2 . back ⟨1, 0⟩];
  back ⟨1, 1⟩ ↔ [-1/2 . exit ⟨0, 1⟩ + -1/2 . exit ⟨1, 1⟩
                + 1/2 . back ⟨0, 1⟩ + 1/2 . back ⟨1, 1⟩]
}

trace kickbackExact on 3/5 . ⟨0, 0⟩ + 4/5 . ⟨1, 1⟩
```

The input is entangled between ancilla and data. Its output is $3|0,1\rangle/5 + 4|1,0\rangle/5$, exactly the result of applying J to the second component. The same identity on arbitrary joint inputs follows from the two-sector calculation above.

F.2 Other retained histories

In the map example of *From Symmetric Pattern-Matching to Quantum Control*, the iso argument $g : a \leftrightarrow b$ is applied to a head and the recursive iso f to the tail. In the source's list notation the two clauses read [22, § 4]

$$[] \leftrightarrow [],$$
$$h : t \leftrightarrow \text{ let } x = g\,h \text{ in let } y = f\,t \text{ in } x : y.$$

If g is Hadamard and the input has length two, every output of the recursion still has length two. A superposition of input lengths two and three likewise produces two components that can be distinguished by length alone. This explains how a reversible program can retain the depth of its recursion without ever measuring it. The cyclic register in (4.1) lets us make the same distinction, up to its period, on a body whose input need not contain a list.

An exact record may also be carried by orthogonal time slots: one tick of delay on the return wire puts $x_0u, x_1u, \ldots$ in consecutive output slots [3, § 3.4]. In the common-clock halting construction, a common post-halt ancilla sequence likewise distinguishes halt times at a fixed observation time. The original model also allows different ancilla sequences across branches, which can restore interference later [17]. Our exact record requires distinct halt times to remain orthogonal at observation. Appendix H gives the corresponding short-pulse and single-frequency experiments.

A case inside or outside feedback. Let θ have $\cos\theta = t = 3/5$ and rotate the guard alone. A phase z on the returning output gives $\theta_0(z) = (t - z)/(1 - tz)$ by summation of the scalar exits. The case $I \oplus \mathrm{trace}(\mathrm{rotbody}(\theta, I); (I \oplus zI))$ on a reference qubit therefore denotes $\mathrm{diag}(1, \theta_0(z))$. If instead the data bit selects the phase on a return *inside* the guard rotation, the two data sectors get $\theta_0(1)$ and $\theta_0(z)$. The result is $\mathrm{diag}(-1, \theta_0(z))$. The bypass in the first program makes no visit to the loop,

whereas both sectors in the second visit it. At $z = i$, the two open bodies are the following literal programs. A factor `(0 :+ 1)` contributes i only to a returning output.

```
round phaseOutside : Qubit + Unit = {
  exit 0 ↔ exit 0;
  exit 1 ↔ [3/5 . exit 1 + (0 :+ 1) . 4/5 . back unit];
  back unit ↔ [-4/5 . exit 1 + (0 :+ 1) . 3/5 . back unit]
}

trace phaseOutside on 3/5 . 0 + 4/5 . 1

round phaseInside : Qubit + Qubit = {
  exit 0 ↔ [3/5 . exit 0 + 4/5 . back 0];
  exit 1 ↔ [3/5 . exit 1 + (0 :+ 1) . 4/5 . back 1];
  back 0 ↔ [-4/5 . exit 0 + 3/5 . back 0];
  back 1 ↔ [-4/5 . exit 1 + (0 :+ 1) . 3/5 . back 1]
}

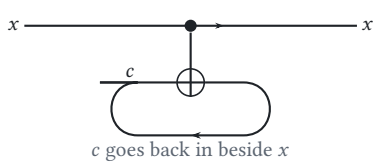
trace phaseInside on 3/5 . 0 + 4/5 . 1
```

Here $\theta_0(i) = (15 - 8i)/17$. On the stated input the first returns $3|0\rangle/5 + (12/17 - 32i/85)|1\rangle$, while the second negates only the $|0\rangle$ coefficient. Thus the position of the case changes the relative phase between the two data alternatives. On $|+\rangle$ followed by H, their probabilities of outcome 1 are $1/17$ and $16/17$, respectively.

G Reading the feedback diagrams

The sheet diagrams read from bottom to top. Each summand has a separate surface. Looking along the depth of one surface reveals the wires of its tensor factors, so two registers on one sheet can be used together, while different sheets carry alternative components. The dark segment at which surfaces meet is a seam of the body. Closing a summand bends its entire surface back to the input. This last bend denotes the trace proved here. The underlying sheet calculus describes sum, tensor and composition [8, Defs. 16, 18–20]. A trace in the sense of this paper closes an entire summand. For instance the ordinary partial matrix trace of controlled-X over the target gives $2|0\rangle\langle 0|$. For the summand trace in Figure 26, reorder the tensor factors to group by the target c, which carries the exit or return tag. The control x remains the external data. Control 0 leaves c unchanged and exits immediately. Control 1 flips c twice and exits after one return. The exit operators are $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$, whose sum is the identity on the control. These computations even have different types of interpretation.

(a) trace for ⊗: a wire fed back beside the others

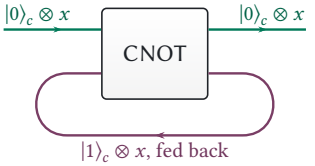


$\sum_k (I \otimes \langle k|)\,\mathrm{CNOT}\,(I \otimes |k\rangle)$
$= |0\rangle\langle 0|\,\mathrm{tr}(I) + |1\rangle\langle 1|\,\mathrm{tr}(\mathrm{X}) = 2|0\rangle\langle 0|$

nothing splits and nothing passes twice;
not unitary

$\mathrm{Tr}^{\mathbb{C}_c^2}_{\otimes}(\mathrm{CNOT}) : \mathbb{C}_x^2 \longrightarrow \mathbb{C}_x^2$

(b) trace for ⊕: the guard splits the state



$A = |0\rangle\langle 0|,\ B = C = |1\rangle\langle 1|,\ D = |0\rangle\langle 0|$ on x
$\mathrm{Tr}^{\oplus}\,\mathrm{CNOT} = A + BC = I$

$x = |1\rangle$ goes round once and leaves;
$\mathrm{Tr}^{Y}_{\oplus}(U) : X \longrightarrow X$ unitary

one controlled flip, x controls c; the guard decomposition reorders the tensor factors

Fig. 26. A tensor-factor partial trace and a summand feedback trace of controlled-X evaluate different diagrams.

G.1 The part of the return space that an input can reach

Figures 27 and 28 show the invariant-space argument and its finite decay bound. There is also a finite way to identify Y_0 without diagonalizing D. Set $K_0 = Y$ and $K_{j+1} = \ker B \cap D^{-1}(K_j)$. Then K_j is the set of vectors with their first j emissions zero. An equality $K_j = K_{j+1}$ makes this space invariant and fixes all subsequent terms; strictly decreasing steps can occur at most $\dim Y$ times. Consequently $K_m = Y_0$ for $m = \dim Y$. When $m > 0$ and $Y_1 \neq 0$, the norm of D_1^m is strictly less than one: a unit vector achieving norm one would have emitted nothing in the first m rounds and would lie in Y_0. Compactness supplies a maximum, hence a uniform geometric bound. This gives another proof of the decay used in Appendix B.

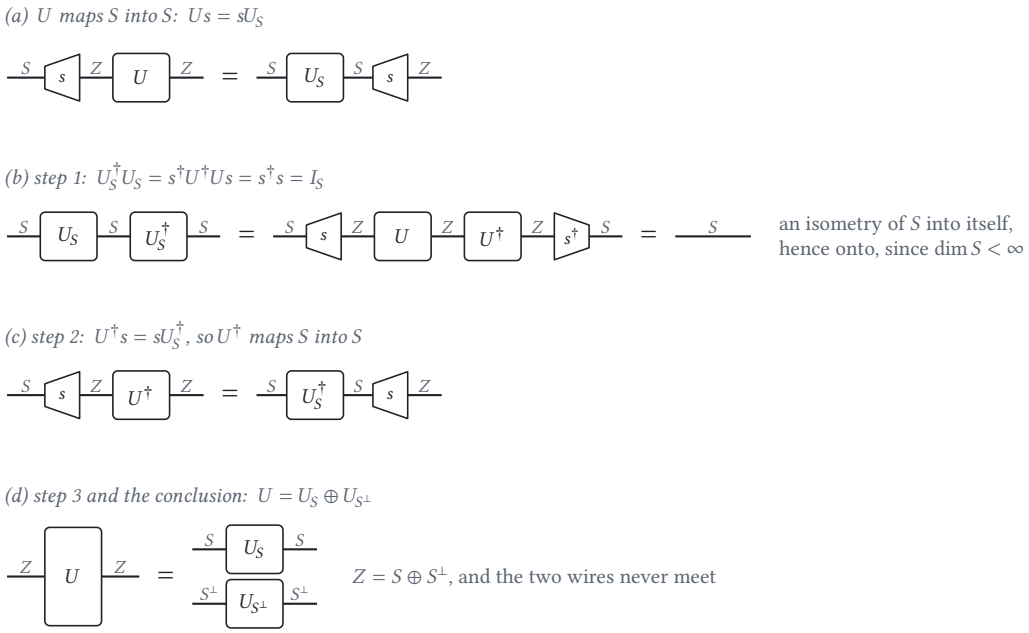


Fig. 27. For a unitary $U : Z \to Z$, let $s : S \rightarrowtail Z$ be the inclusion of a finite invariant subspace. Norm preservation gives an isometry on S, which is onto in finite dimension; hence S reduces U.

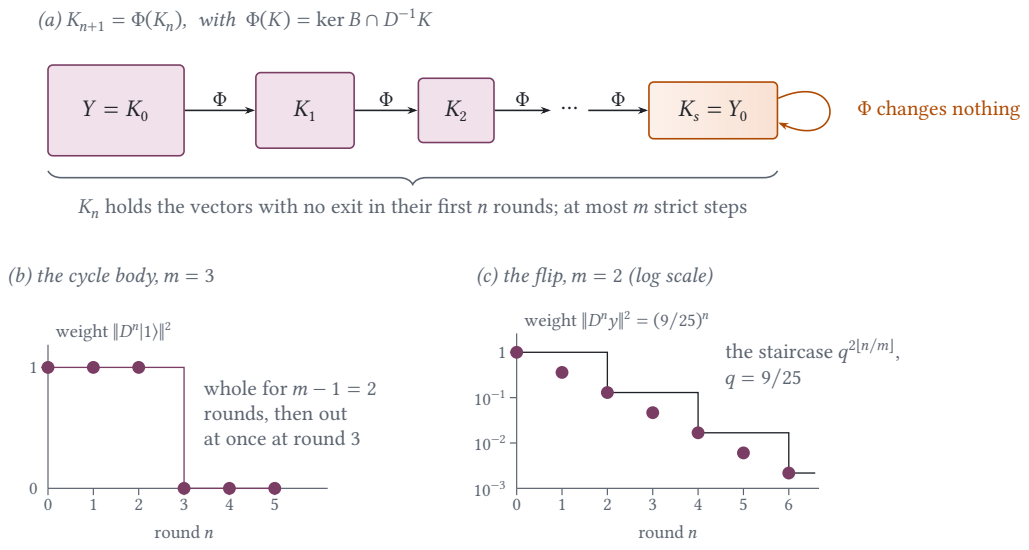


Fig. 28. The no-emission spaces stabilize within $\dim Y$ steps. Panel (b) is the four-state cycle $0 \mapsto 1 \mapsto 2 \mapsto 3 \mapsto 0$, with external state 0. Its three-dimensional return block has $D|1\rangle = |2\rangle$, $D|2\rangle = |3\rangle$ and $D|3\rangle = 0$. In panel (c), $\|D_1\| = 3/5$ and $m = 2$, so $q = \|D_1^2\| = 9/25$. The exact residual weight is $(9/25)^n$, and the staircase bounds it by $q^{2\lfloor n/m \rfloor}$.

G.2 Effects and trace laws

Figure 12 places the three readings in the Π family. Figure 29 adds the partial-injection model and the compact completion. The finite counter is obtained by extending the body before tracing. Figure 30 shows why coherent sum does not descend to channels. The trace on finite isometries agrees on embedded unitaries with the original coherent trace [3, Cor. 3.3.16], so allocation alone does not turn coherent feedback into measurement. The exact countable record is a semantic limit of the finite recording programs. Measuring the tag instead uses the classical sum of operator blocks. All three constructions start from the same iso body.

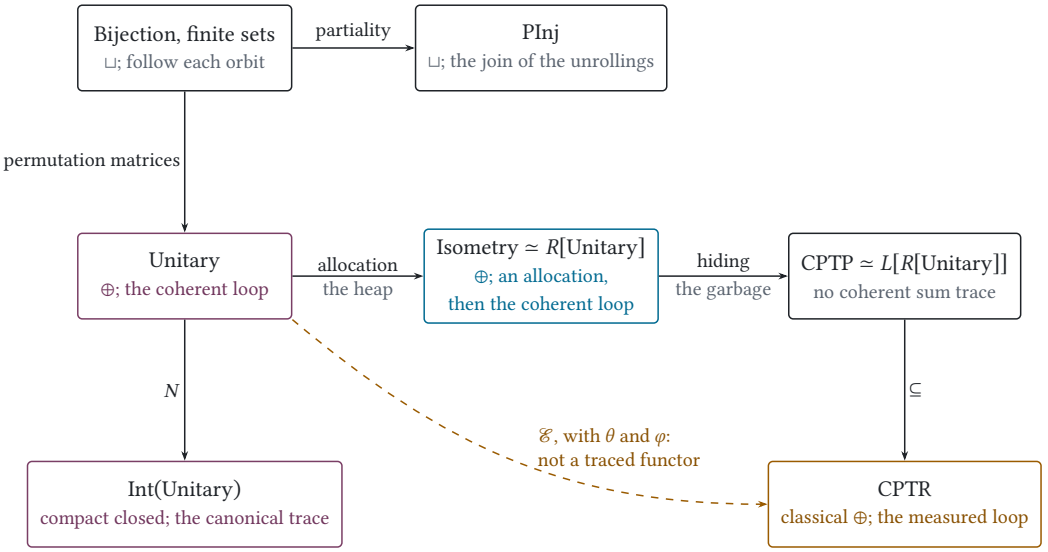


Fig. 29. The measured loop uses a classical sum of operator algebras. The dashed body translation is $U \mapsto \Phi_U = \varphi \circ \mathcal{E}_U \circ \theta$. The direct-sum trace of an embedded unitary remains coherent on the isometry layer.

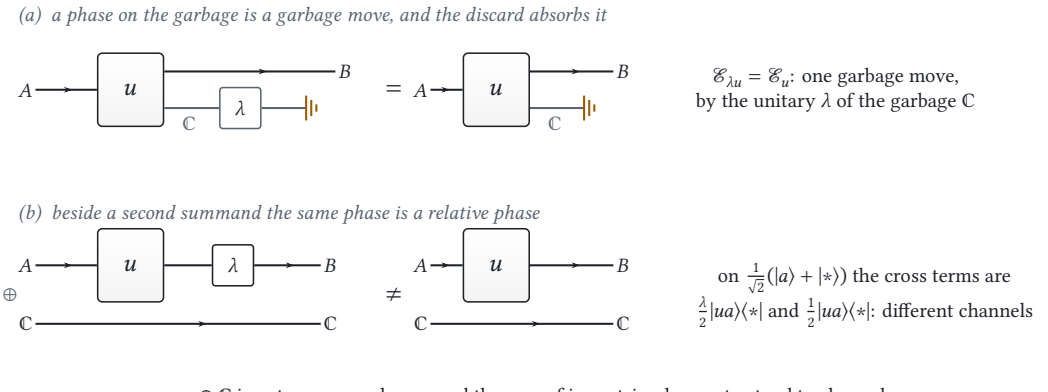


Fig. 30. Here $u : A \rightarrowtail B$ is an isometry, $A \neq 0$, and $|\lambda| = 1$ with $\lambda \neq 1$. A one-dimensional garbage factor absorbs this phase. Putting u in one arm of a coherent sum exposes it instead.

G.3 Composition and the backward wire

The summand matters. In a traced monoidal category the trace closes one component of a morphism $X \odot Y \to X' \odot Y$, for the category's chosen product $\odot$ [14]. Here it is $\oplus$. The usual tensor trace on finite linear maps is a different operation and need not send a unitary to a unitary: tracing the factor $\mathbb{C}^2$ out of $I_X \otimes I_{\mathbb{C}^2}$ gives $2I_X$.

Consider the iso swapping external and internal copies of a (Figure 34). Its input makes one return and comes out unchanged, so its coherent trace is the identity: the law called *yanking*. A K-clock with $K > 1$ distinguishes this path from an immediate exit by labels 1 and 0. Naturality and sliding move external and internal operations across a trace, as drawn in Figure 33 [4, 14]. Figure 12 follows three further equations, with canonical reassociations understood. Closing two return sheets successively is the same as closing their sum. A bypass stays outside the closure. In sequence, the first loop's external output supplies the second, but each return stays with its own body. Writing their exit families as x_i^U, x_j^V, the combined body emits $\sum_{i+j=k} x_j^V x_i^U$ at total count k. These are equations of coherent feedback at the displayed external types.

Writing the guard as a tensor factor and applying distributivity gives exactly the two injection tags in a clause (Figure 31). Naturalities of the additive trace apply to this representation. Closing the body $I \oplus U$ over U's return port gives $I \oplus L$: the immediate exit is $I \oplus x_0$ and the later exits are $0 \oplus x_k$. Closing disjoint internal ports successively gives the simultaneous solution by the argument of Appendix B.1, as drawn in Figure 32. For two loops in sequence, at a common phase z their responses multiply as $\Theta_2(z)\Theta_1(z)$, because the external output of the first enters the second and their return ports are separate. Expanding the product gives the convolution $\sum_{j=0}^{k} x_{k-j}^{(2)} x_j^{(1)}$ at total count k. The adjoint of a coherent loop is the loop of the adjoint body: its blocks give exactly $x_k^\dagger$ as exits and the series is absolutely convergent (Figure 35).

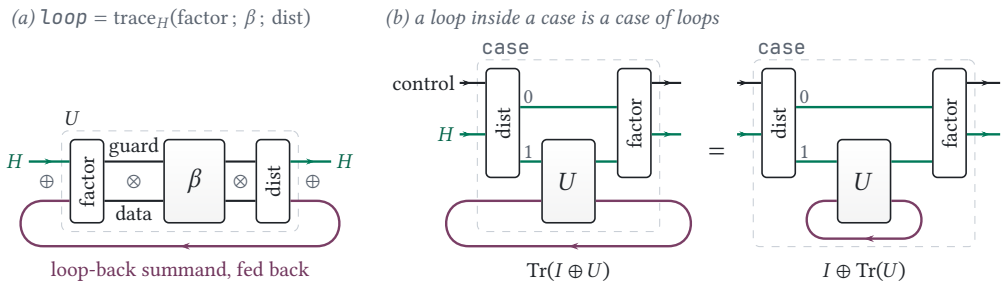


Fig. 31. Distributivity makes the guard a summand. A case with a separate untouched arm distributes over the trace of the other arm.

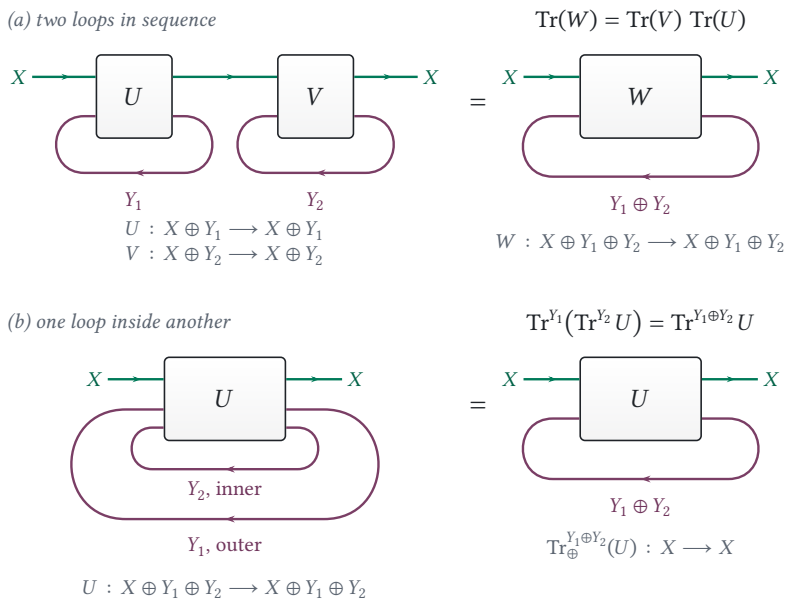
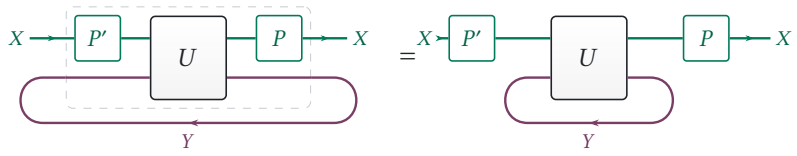


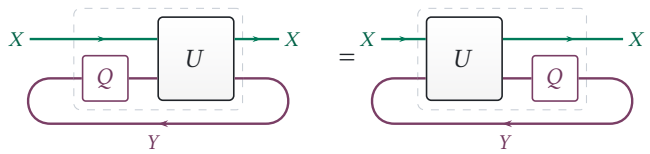
Fig. 32. Sequencing convolves the exit operators; nested closure over internal ports solves their feedback equations jointly.

(a) naturality



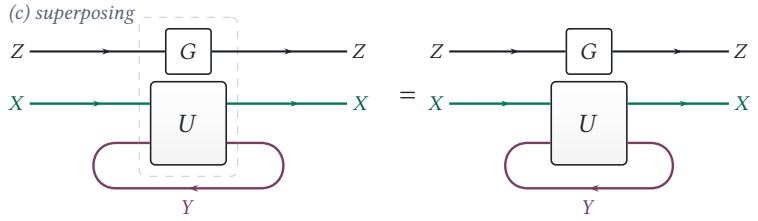
$$\mathrm{Tr}^{Y}\big((P \oplus I)\,U\,(P' \oplus I)\big) = P\ \mathrm{Tr}^{Y}(U)\,P'$$

(b) sliding



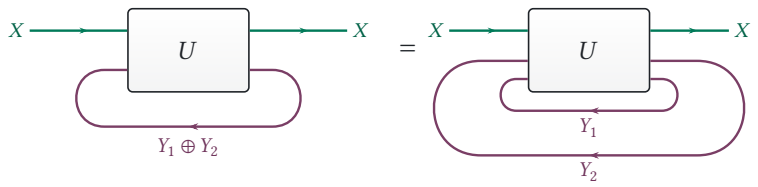
$$\mathrm{Tr}^{Y}\big(U\,(I \oplus Q)\big) = \mathrm{Tr}^{Y}\big((I \oplus Q)\,U\big)$$

(c) superposing



$\mathrm{Tr}^{Y}(G \oplus U) = G \oplus \mathrm{Tr}^{Y}(U)$, the loop taken over Y

(d) vanishing



$$\mathrm{Tr}^{Y_1 \oplus Y_2}(U) = \mathrm{Tr}^{Y_2}\big(\mathrm{Tr}^{Y_1}(U)\big)$$

Fig. 33. Naturality, sliding, superposing and vanishing are typed equations for coherent loops. In panel (a), $P, P' : X \to X$.

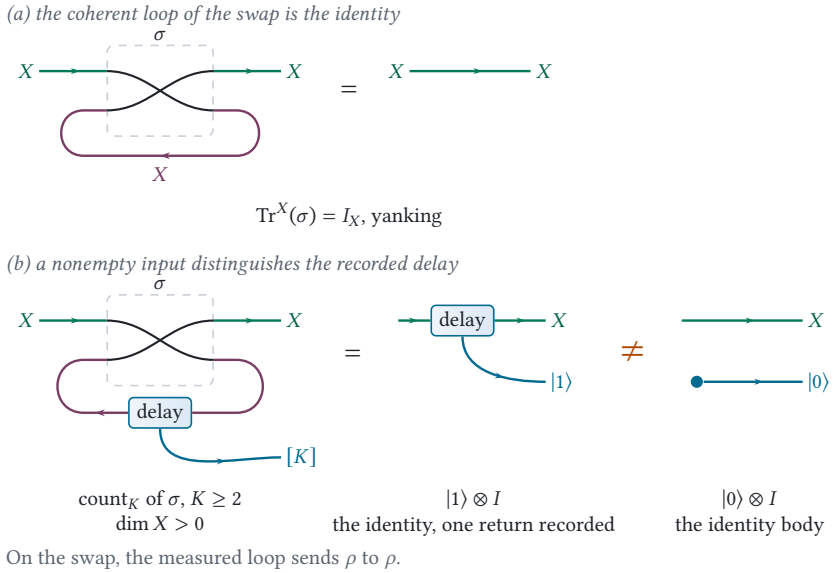


Fig. 34. The swap is between two copies of X. Its coherent trace returns the input. For $\dim X > 0$ and $K \geq 2$, a recording loop distinguishes one return from an immediate exit.

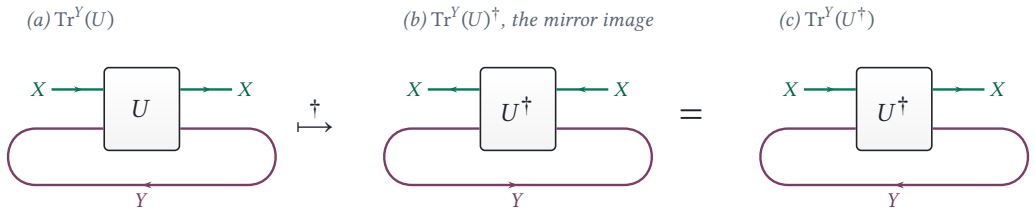


Fig. 35. Reversing a coherent body exchanges entry and exit and takes the adjoint of each exit amplitude.

In the loop diagram a wire runs backwards from the returning output to the next input. To make this drawing into a typed composition, the Int construction pairs each forward type P^+ with a backward type P^-. A map from P to Q can then consume P^+ from its left and Q^- from its right, so its underlying unitary has type $P^+ \oplus Q^- \to Q^+ \oplus P^-$. Figure 37 draws these ports. Composition connects the middle ports by feedback, as in Figure 38. In particular, an ordinary space enters the construction as $N(X) = (X, 0)$, with no backward input. Bending Y around an embedded body identifies the amplitude leaving it, $Cu + Dy$, with the amplitude y supplied on the same wire. The compact closed trace drawn in Appendix G, Figure 36, therefore returns exactly $N(L)$, by the very wire equation used in Theorem B.1 [14] [18, Thms. 5.15–5.16]. Inserting a successor changes what comes back along that wire. Measuring the tag changes what the wire carries from an amplitude to a classical alternative.

The sum in Int reverses the order of negative components:

$$(P^+, P^-) \oplus (Q^+, Q^-) = (P^+ \oplus Q^+, Q^- \oplus P^-).$$

The dual of $(Y, 0)$ is $(0, Y)$, and the unit and counit are identity wires on Y with their ports rearranged. Composing two maps plugs the forward and backward middle ports together and takes the original

trace. With $\eta_{NY} : 0 \to (-Y) \oplus NY$ and $\varepsilon_{NY} : NY \oplus (-Y) \to 0$, the typed closure is

$$N(\mathrm{Tr}^Y U) = (I_{NX'} \oplus \varepsilon_{NY})(NU \oplus I_{-Y})(I_{NX} \oplus \sigma_{-Y,NY})(I_{NX} \oplus \eta_{NY}).$$

The swap puts the cup's two outputs in the order required by $NU \oplus I_{-Y}$. Its underlying map is $I_{Y \oplus Y}$, since both source and target pairs are (Y, Y). With the external identity this is $I_{X \oplus Y \oplus Y}$, explaining the second 1 in Figure 36(b). The four regions between the three dashed slices in Figure 36(b) show these four factors in execution order. If two perfect reflectors enclose a mode, that mode need not have a unique internal amplitude, but it has a unique external response by Theorem B.1. A finite clock bends the enlarged internal type $[K] \otimes Y$ and advances its label on each passage. These statements require no tensor-product duals of unitary gates themselves. Figure 39 separates these constructions. Under direct sum a cup from 0 to $Y \oplus Y^*$ cannot be unitary unless $Y = 0$. Under tensor, a Hilbert-space cup and cap obeying the snake identity on an n-dimensional object have product of norms at least n: closing to a scalar gives n and Cauchy–Schwarz bounds it by the product. Thus for $n \geq 2$ they cannot both be contractions. The compact completion uses the sum, where its objects are pairs.

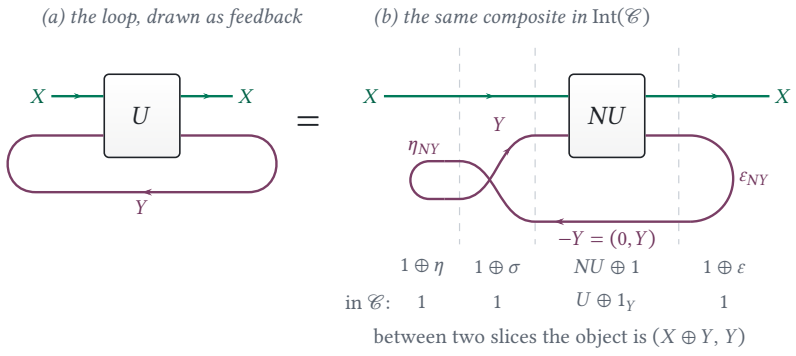


Fig. 36. For $X' = X$ the negative port in the compact closed completion connects the output of the return summand to its input. The same amplitude must satisfy the feedback equation at both ends.

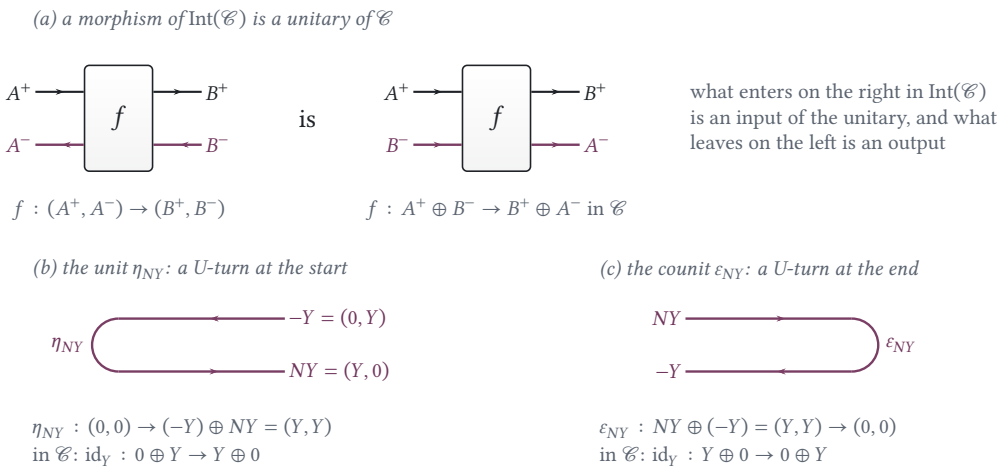
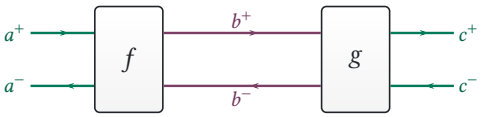


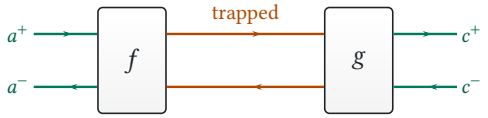
Fig. 37. Objects in the compact completion have forward and backward components. Cups and caps bend a port within that completion.

(a) $g \circ f$: a loop over the middle wires



each round trip from f to g and back
multiplies by $f_{12}\, g_{21}$; the composite adds them

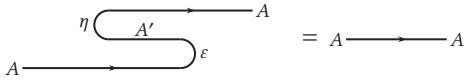
(b) two perfect mirrors: $f_{12}\, g_{21} = I$



the light between them is a dead part;
$g \circ f$ is the outer reflection of each mirror

Fig. 38. With rows and columns in the displayed order, $f_{12} : B^- \to B^+$ and $g_{21} : B^+ \to B^-$. Thus $b^+ = f_{11}a^+ + f_{12}b^-$ and $b^- = g_{21}b^+ + g_{22}c^-$ give round trip $f_{12}g_{21}$ on B^+. A perfectly reflecting interior is a dead summand invisible from outside.

(a) the tensor product: a dual would make the loop of the identity the dimension

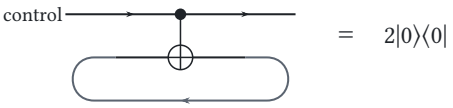


the snake equation defines the dual



so $\varepsilon\, \sigma\, \eta = \dim A$, while $|\varepsilon\, \sigma\, \eta| \le \|\varepsilon\|\, \|\eta\| \le 1$
among contractions: no dual once $n \ge 2$

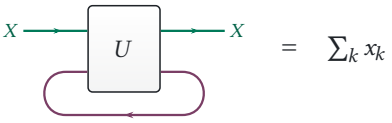
(b) the trace of a factor adds the blocks of one application



tr over the target, a factor

$(I \otimes \langle 0|)\,\mathrm{CNOT}\,(I \otimes |0\rangle) + (I \otimes \langle 1|)\,\mathrm{CNOT}\,(I \otimes |1\rangle)$
$= |0\rangle\langle 0| + |0\rangle\langle 0|$, not a contraction

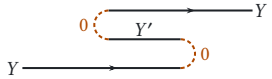
(c) the trace of a summand adds the exits of the rounds



the loop-back summand Y

$y = Cu + Dy$ implies
$U(u, y) = (\mathrm{Tr}^Y(U)u, y)$,
so $\|\mathrm{Tr}^Y(U)u\| = \|u\|$

(d) the direct sum: the unit and counit through the zero space are zero maps

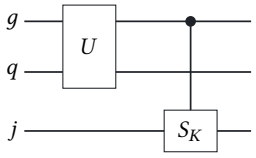


not the identity of Y: every map into or out of the zero space is zero,
and a unitary into or out of it exists only when $Y = 0$

Fig. 39. The compact completion supplies the negative type. The original tensor product on contractions does not supply a dual of dimension two or greater.

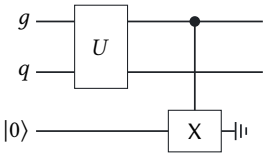
Figure 40 puts the counter increment and the tag-copying step beside the same body.

Keep the count modulo K



$S_K|j\rangle = |j + 1 \bmod K\rangle$.

Measure only the guard tag



Discard the fresh bit after this call.

Fig. 40. One call to the same body U, followed by an operation controlled on the outgoing guard $g = 1$. A filled dot denotes a coherent control. The left circuit carries j into the next round. In the right circuit the discarded bit distinguishes the tags without distinguishing states of the payload q. Bare crossings have no gate.

G.4 The same operations on sheets

The finite counter in Figure 6 uses the depth of the page. The successor acts on its counter wire and leaves the payload wire alone. Figures 8 and 10 place the measurement and phase on that return. Figure 12 shows how separate return sheets compose, in the same upward orientation. In particular a seam labeled U acts on its incident summands even when one drawing of the external wire is partially hidden by the returning surface.

H A ring realization of the same exits

The reference circuit in Section 3 makes the scalar phase observable. The same exits arise from an ideal lossless coupler between an external waveguide mode and a circulating ring mode. Its through amplitude is $t = 3/5$ and its crossing amplitude is, where $s = 4/5$. Figure 41 gives its clauses and feedback equation.

The coupler is $(1 \oplus i)R_\theta(1 \oplus (-i))$. Changing the returning basis in this way gives entrance and exit coefficients both equal to is, with return coefficient t. Their product on exit $k + 1$ is $(is)t^k(is) = -s^2t^k$, so every external exit is unchanged. With phase z per circuit, its single-frequency response is therefore $\Theta_{\text{ring}}(z) = (t - z)/(1 - tz)$, as for the rotation. The same coupler has a literal rational-complex implementation:

```
round ringCoupler : Unit + Unit = {
  exit unit ↔ [3/5 . exit unit + (0 :+ 1) . 4/5 . back unit];
  back unit ↔ [(0 :+ 1) . 4/5 . exit unit + 3/5 . back unit]
}

trace ringCoupler on unit
```

The command returns $-|*\rangle$, as does the guard rotation. The returning basis change preserves every exit, not only their sum.

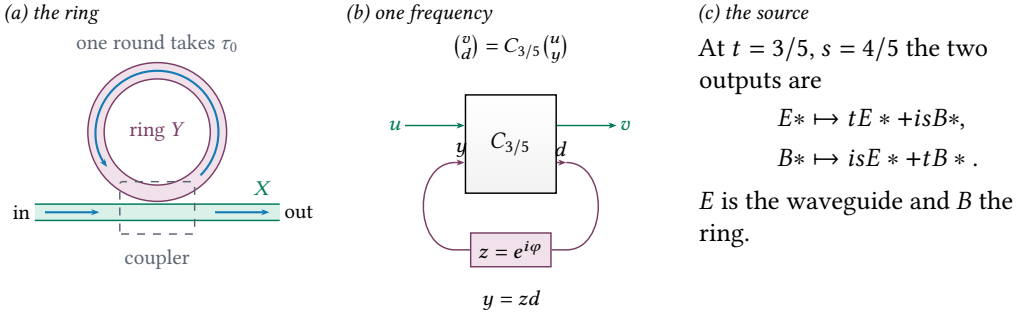


Fig. 41. The ring, the feedback equation at a single frequency, and the two clauses of the coupler. Here d is the amplitude leaving the coupler inside the ring, and $y = zd$ enters it again.

In this lossless coupler [2, Eqs. (6)–(7)], the observation depends on the input pulse. If the pulse lasts less than one circuit time, the outgoing pulse on each round ends before the next begins. Measuring the arrival time then distinguishes the exits and gives their probabilities. At a single frequency all the returning waves occupy the same mode and their amplitudes add [21, Eqs. (32)–(33)]. Figure 42 draws these two mode specifications.

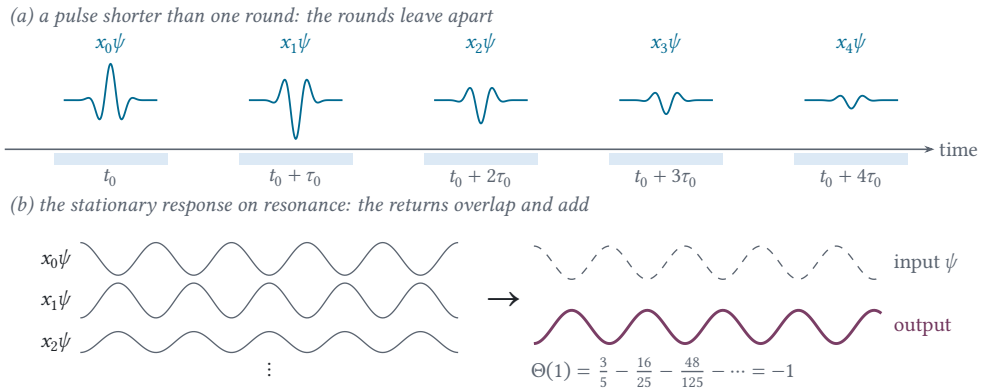


Fig. 42. A short pulse leaves in distinct time bins, with exit amplitudes $x_k u$. At one frequency the same exits contribute to one outgoing mode and add coherently.

Putting the bypass beside the ring gives the exits of `bypassRound`. Prepare its external qubit in $|+\rangle$, run the two paths, and apply H to compare them. At a frequency for which one trip around the ring brings the wave back in phase with itself, the round-trip factor is $z = 1$. This is resonance. The ring response is then -1 and the difference port has probability $|1 - (-1)|^2/4 = 1$. For a short pulse the first time bin contributes $(1 - t)^2/4$ to this port, and the later bins contribute their probabilities separately:

$$\Pr(b = 1) = \frac{1}{4}\left((1 - t)^2 + \sum_{k \geq 1} s^4 t^{2(k-1)}\right) = \frac{1}{4}\left((1 - t)^2 + 1 - t^2\right) = \frac{1}{5}.$$

These are the coherent and measured predictions from Section 4. The scalar ring passes agreement, but the two-arm body fails: its immediate exit contains the bypass and its later exits do not. Forming that bypass uses a coherent sum of isos before channelization, an operation unavailable to a channel context.

The two experiments specify different input and output modes, one frequency or orthogonal time slots. A finite wave packet and observation window require their own specification before these predictions can be compared.

I Observing the response

For a fixed $|z| = 1$, the phase iso $P_z^c = \{s \leftrightarrow z \cdot s\}_{s \in \mathcal{B}(c)} : c \leftrightarrow c$ gives the marked source body $\beta; (I_b \oplus P_z^c)$. The bypass caller is

$$H; \mathsf{trace}_1\big(\mathsf{bypassRound}; (I_{\mathbf{Qubit}} \oplus P_z^1)\big); H.$$

Thus the phase variation used in the body of the paper is an ordinary finite iso construction.

The controlled body. For $|z| = 1$ and $\delta \in \mathbb{R}$, let the reference space be $H_r = \mathbb{C}^2$, and put $P_\delta = \mathrm{diag}(1, e^{i\delta})$. Use distributivity to identify $H_r \otimes (X \oplus Y)$ with $(H_r \otimes X) \oplus (H_r \otimes Y)$. The body in Figure 11 is the finite unitary

$$W_{z,\delta} = \big(I_{H_r \otimes X'} \oplus (P_\delta \otimes zI_Y)\big)(I_{H_r} \otimes U).$$

It has external type $H_r \otimes X \longrightarrow H_r \otimes X'$ and returning summand $H_r \otimes Y$. For reference basis value $r = 0$ its returning phase is z, and for $r = 1$ it is $ze^{i\delta}$. It never changes that basis value. Its kth exit on the two sectors is therefore respectively $z^k x_k$ and $(ze^{i\delta})^k x_k$. The closed gate is

$$|0\rangle\langle 0|_r \otimes \Theta(z) + |1\rangle\langle 1|_r \otimes \Theta(ze^{i\delta}).$$

For reference input $|+\rangle$ and unit data u, the closed gate returns

$$\frac{|0\rangle_r \otimes \Theta(z)u + |1\rangle_r \otimes \Theta(ze^{i\delta})u}{\sqrt{2}}.$$

Applying $S^\dagger$ and H gives data vectors on reference outcomes $0, 1$: $(\Theta(z)u - i\Theta(ze^{i\delta})u)/2$ and $(\Theta(z)u + i\Theta(ze^{i\delta})u)/2$. Their squared-norm difference is $\mathrm{Im}\langle\Theta(z)u, \Theta(ze^{i\delta})u\rangle$. Taylor expansion at $\delta = 0$ gives the displayed derivative readout in Figure 11.

Thus the straight reference line in the drawing denotes a tensor factor carried on both alternative sheets. Its controlled phase acts on each return. No coherent control of a channel hole is used.

At $z = 1$, this construction compares $\Theta(1)u$ and $\Theta(e^{i\delta})u$. For the scalar response $(3-5z)/(5-3z)$, choose $\delta = \pi/2$. The imaginary overlap is $8/17$, so the two outcome probabilities are $25/34$ and $9/34$. This finite-phase readout is the exact overlap, whereas the derivative formula is its first-order expansion at $\delta = 0$. This specializes Figure 11 to the controlled body $W_{1,\delta}$.